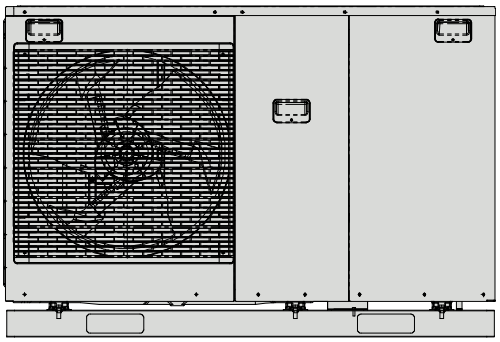
Order No: PAPAMY2003034CE

Service Manual

Air-to-Water Heatpump



Mono Bloc Unit
WH-MDC05J3E5
WH-MDC07J3E5
WH-MDC09J3E5

Destination
Europe
Turkey

⚠ WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public.
It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product.
Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the products dealt with in this service information by anyone else could result in serious injury or death.

⚠ CAUTION

R32 REFRIGERANT – This Air Conditioner contains and operates with refrigerant R32.
THIS PRODUCT MUST ONLY BE INSTALLED OR SERVICED BY QUALIFIED PERSONNEL.
Refer to National, State, Territory and local legislation, regulations, codes, installation & operation manuals, before the installation, maintenance and/or service of this product.

⚠ PRECAUTION OF LOW TEMPERATURE

In order to avoid frostbite, be assured of no refrigerant leakage during the installation or repairing of refrigerant circuit.

Panasonic®

© Panasonic Corporation 2020.

TABLE OF CONTENTS

PAGE

1. Safety Precautions 4

2. Precaution for Using R32 Refrigerant 6

3. Specifications 10

3.1 WH-MDC05J3E5 10
3.2 WH-MDC07J3E5 13
3.3 WH-MDC09J3E5 16

4. Features 19

5. Location of Controls and Components 20

5.1 Indoor Unit 20

6. Dimensions 38

6.1 Mono Bloc Unit 38

7. Refrigeration and Water Cycle Diagram 39

7.1 Standard System (Optional PCB not connected) 40
7.2 Advance System (Optional PCB connected) 40

8. Block Diagram 41

8.1 WH-MDC05J3E5 41
8.2 WH-MDC07J3E5 WH-MDC09J3E5 42

9. Wiring Connection Diagram 43

9.1 Water System 43
9.2 Refrigerant System 44

10. Electronic Circuit Diagram 46

10.1 Water System 46
10.2 Refrigerant System 47

11. Printed Circuit Board 49

11.1 Water System 49
11.2 Refrigerant System 51

12. Installation Instruction 52

12.1 Mono Bloc Unit 52
12.2 Variation of system 64
12.3 How to Fix External Device 71
12.4 System Installation 74
12.5 Service and maintenance 85

13. Installation and Servicing Air Conditioner using R32 86

13.1 About R32 Refrigerant 86
13.2 Characteristics of R32 Refrigerant 86
13.3 Refrigerant piping installation • Tools used in services 88
13.4 New installation, Relocation, Repairing of Refrigerant Cycle System The Procedures 92
13.5 Piping installation of R32 93
13.6 Installation, Relocation, and Service 94
13.7 Repairing of refrigerant cycle / Brazing point 98

PAGE

13.8 <Reference> Analysis method for no error code, no cooling / no warming 104

14. Operation and Control 106

14.1 Basic Function 106
14.2 Water Pump 116
14.3 Indoor Unit Safety 119
14.4 Auto Restart Control 119
14.5 Indication Panel 119
14.6 Indoor Back-Up Heater Control 120
14.7 Tank Heater Control 121
14.8 Base Pan Heater Control (Optional) 122
14.9 Force Heater Mode 123
14.10 Powerful Operation 124
14.11 Quiet Operation 125
14.12 Sterilization Mode 125
14.13 Outdoor Ambient Thermo OFF Control 126
14.14 Alternative Outdoor Ambient Sensor Control 126
14.15 Force DHW mode 127
14.16 SMART DHW mode 127
14.17 DHW Capacity Setting 128
14.18 Anti Freeze Control 128
14.19 Solar Operation (Optional) 129
14.20 Boiler Bivalent Control 131
14.21 External Room Thermostat Control (Optional) 133
14.22 Three Ways Valve Control 134
14.23 Two Ways Valve Control 134
14.24 External OFF/ON Control 135
14.25 External Compressor Switch (Optional PCB) 136
14.26 Heat/Cool Switch (Optional PCB) 137
14.27 SG Ready Control (Optional PCB) 137
14.28 Demand Control (Optional PCB) 139
14.29 Holiday Mode 139
14.30 Dry Concrete 140
14.31 Flow Sensor 140

15. Protection Control (WH-MDC05J3E5) 141

15.1 Protection Control for All Operations 141
15.2 Protection Control for Heating Operation 143
15.3 Protection Control for Cooling Operation 144

16. Protection Control (WH-MDC07J3E5 WH-MDC09J3E5) 145

16.1 Protection Control for All Operations 145
16.2 Protection Control for Heating Operation 147
16.3 Protection Control for Cooling Operation 148

17. Servicing Mode 149

17.1 Expansion Vessel Pre Pressure Checking 149
17.2 Maintenance 149
17.3 Pump Down Procedures 150
17.4 How To Adjust Pump Speed 151

17.5 How To Adjust Pump Speed 152
17.6 How To Unlock Cool Mode.....153
17.7 EEPROM Factory Default Data Setup Procedure 154
17.8 Dry Concrete Setup 155

18. Maintenance Guide 157

18.1 Maintenance Menu 158

19. Troubleshooting Guide.....160

19.1 Refrigeration Cycle System 160
19.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current.....161
19.3 Breakdown Self Diagnosis Function.....162
19.4 Error Codes Table 164
19.5 Self-diagnosis Method 166

20. Disassembly and Assembly Instructions 214

20.1 To Remove The Cabinet Front Plate.....214
20.2 To Remove The Cabinet Top Plate.....214
20.3 To Remove The Cabinet Rear Plate 215
20.4 To Remove Pressure Gauge.....215
20.5 To Remove RCCB.....216
20.6 To Remove Reactor 216
20.7 To Remove Water System Electronic Controller Board.....217
20.8 To Remove Refrigerant Side Control Board Complete.....218
20.9 To Remove Pressure Relief Valve and Air Purge Valve 219
20.10 To Remove Flow Sensor 219
20.11 To Remove Water Pump 220
20.12 To Remove Heater 221
20.13 To Remove Filter 222

21. Technical Data.....223

21.1 Operation Characteristics 223
21.2 Heating Capacity Table 235
21.3 Cooling Capacity Table 236

22. Exploded View and Replacement Parts List.....237

22.1 Water System 237
22.2 Refrigerant System.....238

• Specifications, designs and contents in this Service Manual are subject to change without notice.

1. Safety Precautions

- Read the following “SAFETY PRECAUTIONS” carefully before installation of (Mono bloc) Air-to-Water Heatpump system (hereafter referred to as “Mono bloc unit”).
- Electrical works and water installation works must be done by licensed electrician and licensed water system installer respectively. Be sure to use the correct rating and main circuit for the model to be installed.
- The caution items stated here must be followed because these important contents are related to safety. The meaning of each indication used is as below.
  Incorrect installation due to ignorance or negligence of the instructions will cause harm or damage, and the seriousness is classified by the following indications.

⚠ **WARNING**	This indication shows the possibility of causing death or serious injury.
⚠ **CAUTION**	This indication shows the possibility of causing injury or damage to properties only.

The items to be followed are classified by the symbols:

⃠	Symbol with white background denotes item that is PROHIBITED from doing.
❗ ⏚	Symbol with dark background denotes item that must be carried out.

- Carry out test run to confirm that no abnormality occurs after the installation. Then, explain to user the operation, care and maintenance as stated in instructions. Please remind the customer to keep the operating instructions for future reference.
- If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.

	⚠ **WARNING**	
1.	Do not install Mono bloc unit near handrail of veranda. When installing Mono bloc unit at veranda of high rise building, child may climb up to Mono bloc unit and cross over the handrail and causing accident.	⃠
2.	Do not use unspecified cord, modified cord, join cord or extension cord for power supply cord. Do not share the single outlet with other electrical appliances. Poor contact, poor insulation or over current will cause electrical shock or fire.	⃠
3.	Do not tie up the power supply cord into a bundle by band. Abnormal temperature rise on power supply cord may happen.	⃠
4.	Do not insert your fingers or other objects into the unit, high speed rotating fan may cause injury.	⃠
5.	Do not sit or step on the unit, you may fall down accidentally.	⃠
6.	Keep plastic bag (packaging material) away from small children, it may cause suffocation.	⃠
7.	Do not use pipe wrench to install refrigerant pipe. It might deform the piping and cause the unit to malfunction.	⃠
8.	Do not purchase unauthorized electrical parts for installation, service, maintenance and etc.. They might cause electrical shock or fire.	⃠
9.	This unit is a multi supply appliances. All circuits must be disconnected before accessing to the unit terminals.	⃠
10.	Do not modify the wiring of Mono bloc unit for installation of other components (i.e. heater, etc). Overloaded wiring or wire connection points may cause electrical shock or fire.	⃠
11.	Do not add or replace refrigerant other than specified type. It may cause product damage, burst and injury etc.	⃠
12.	For electrical work, follow the local national wiring standard, regulation and this installation instruction. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in electrical work, it will cause electrical shock or fire.	❗
13.	For water circuit installation work, follow to relevant European and national regulations (including EN61770) and local plumbing and building regulation codes.	❗
14.	Must engage an authorized dealer or specialist for installation. If installation is defective, it will cause water leakage, electrical shock or fire.	❗
15.	Install according to this installation instructions strictly. If installation is defective, it will cause water leakage, electrical shock or fire.	❗
16.	Only use the supplied or specified installation parts. Else, it may causes Mono bloc unit vibrate, fall, water leakage, electrical shock or fire.	❗
17.	Install at a flat, strong and firm location which is able to withstand the Mono bloc unit’s weight. If the location is slanting, or strength is not enough the set will fall and cause injury.	❗
18.	This equipment is strongly recommended to be installed with Residual Current Device (RCD) on-site according to the respective national wiring rules or country-specific safety measures in terms of residual current.	❗
19.	The unit is only for use in a closed water system. Utilization in an open water system may lead to excessive corrosion of the water piping and risk of incubating bacteria colonies, particularly Legionella, in water.	❗
20.	If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.	❗
21.	Select a location where in case of water leakage, the leakage will not cause damage to other properties.	❗

⚠ WARNING	
22.	When installing electrical equipment at wooden building of metal lath or wire lath, in accordance with electrical facility standard, no electrical contact between equipment and building is allowed. Insulator must be installed in between.
23.	This installation may be subjected to building regulation approval applicable to respective country that may require to notify the local authority before installation.
24.	Any work carried out on the Mono bloc unit after removing the front panel which is secured by screws, must be carried out under the supervision of authorized dealer and licensed installation contractor.
25.	This unit must be properly earthed, the electrical earth must not be connected to a gas pipe, water pipe, the earth of a lightning rod or a telephone. Otherwise there is a danger of electrical shock in the event of an insulation breakdown or electrical earth fault in the Mono bloc unit.

⚠ CAUTION	
1.	Do not install the Mono bloc unit in areas where there is a risk of flammable gas leakage. There is a risk of fire if flammable gas accumulates near or around the Mono bloc unit.
2.	Do not release refrigerant during piping work for installation, re-installation and during repairing a refrigeration parts. Take care of the liquid refrigerant, it may cause frostbite.
3.	Make sure the power supply cord does not contact with hot part (i.e. water piping). High temperature may cause insulator of power supply cord damage hence electrical shock or fire.
4.	Do not touch the sharp aluminium fin, sharp parts may cause injury.
5.	Do not apply excessive force to water pipes that may damage the pipes. If water leakage occurs, it will cause flooding and damage to other properties.
6.	Carry out drainage piping as mentioned in installation instructions. If drainage is not perfect, water leakage may happen and may cause damage to properties of the user.
7.	The piping installation work must be flushed before the Mono bloc unit is connected to remove contaminants. Contaminants may damage the Mono bloc unit components.
8.	Select an installation location where it is accessible for maintenance.
9.	Power supply connection to Mono bloc unit. • Power supply point should be in easily accessible place for power disconnection in case of emergency. • Must follow local national wiring standard, regulation and this installation instruction. • Strongly recommended to make permanent connection to a circuit breaker. It must be a double pole switch with a minimum 3.0 mm gap. - Power supply 1: Use approved 30A circuit breaker - Power supply 2: Use approved 30A circuit breaker
10.	Ensure the correct polarity is maintained throughout all wiring. Otherwise, it will cause electrical shock or fire.
11.	After installation, the installer is obliged to verify correct operation of the Mono bloc unit. Check the connection point for water leakage during test run. If leakage occurs, it will cause damage to other properties.
12.	Installation work. Four or more people are required to carry out the installation work. The weight of Mono bloc unit might cause injury if carried by less than four people.

2. Precaution for Using R32 Refrigerant

- The basic installation work procedures are the same as conventional refrigerant (R410A, R22) models. However, pay careful attention to the following points:

	⚠ WARNING
1.	Since the working pressure is higher than that of refrigerant R22 models, some of the piping and installation and service tools are special. Especially, when replacing a refrigerant R22 model with a new refrigerant R32 model, always replace the conventional piping with the R32 and R410A piping on the outdoor unit side. For R32 and R410A, the same pipe can be used.
2.	The mixing of different refrigerants within a system is prohibited. Models that use refrigerant R32 and R410A have a different charging port thread diameter to prevent erroneous charging with refrigerant R22 and for safety. Therefore, check beforehand. [The charging port thread diameter for R32 and R410A is 12.7 mm (1/2 inch).]
3.	Ensure that foreign matter (oil, water, etc.) does not enter the piping.
4.	Operation, maintenance, repairing and refrigerant recovery should be carried out by trained and certified personnel in the use of flammable refrigerants and as recommended by the manufacturer. Any personnel conducting an operation, servicing or maintenance on a system or associated parts of the equipment should be trained and certified.
5.	Any part of refrigerating circuit (evaporators, air coolers, AHU, condensers or liquid receivers) or piping should not be located in the proximity of heat sources, open flames, operating gas appliance or an operating electric heater.
6.	The user/owner or their authorized representative shall regularly check the alarms, mechanical ventilation and detectors, at least once a year, where as required by national regulations, to ensure their correct functioning.
7.	A logbook shall be maintained. The results of these checks shall be recorded in the logbook.
8.	In case of ventilations in occupied spaces shall be checked to confirm no obstruction.
9.	Before a new refrigerating system is put into service, the person responsible for placing the system in operation should ensure that trained and certified operating personnel are instructed on the basis of the instruction manual about the construction, supervision, operation and maintenance of the refrigerating system, as well as the safety measures to be observed, and the properties and handling of the refrigerant used.
10.	The general requirement of trained and certified personnel are indicated as below: a) Knowledge of legislation, regulations and standards relating to flammable refrigerants; and, b) Detailed knowledge of and skills in handling flammable refrigerants, personal protective equipment, refrigerant leakage prevention, handling of cylinders, charging, leak detection, recovery and disposal; and, c) Able to understand and to apply in practice the requirements in the national legislation, regulations and Standards; and, d) Continuously undergo regular and further training to maintain this expertise.
11.	Protect the refrigerating system from accidental rupture due to moving furniture or reconstruction activities.
12.	To ensure no leaking, field-made refrigerant joints indoors shall be tightness tested. The test method shall have a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.15MPa). No leak shall be detected.

⚠ CAUTION

1. Installation (Space)
   - Must comply with national gas regulations, state municipal rules and legislation. Notify relevant authorities in accordance with all applicable regulations.
   - Must ensure mechanical connections be accessible for maintenance purposes.
   - In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.
   - When disposal of the product, do follow to the precautions in #12 and comply with national regulations.
   - Always contact to local municipal offices for proper handling.

2. Servicing

2-1. Service personnel
- Any qualified person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognised assessment specification.
- Servicing shall only be performed as recommended by the equipment manufacturer. Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.
- Servicing shall be performed only as recommended by the manufacturer.
- The system is inspected, regularly supervised and maintained by a trained and certified service personnel who is employed by the person user or party responsible.
- Ensure refrigerant charge not to leak.

2-2. Work
- Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimised.
  For repair to the refrigerating system, the precautions in #2-2 to #2-8 must be followed before conducting work on the system.
- Work shall be undertaken under a controlled procedure so as to minimize the risk of a flammable gas or vapor being present while the work is being performed.
- All maintenance staff and others working in the local area shall be instructed and supervised on the nature of work being carried out.
- Avoid working in confined spaces. Always ensure away from source, at least 2 meter of safety distance, or zoning of free space area of at least 2 meter in radius.
- Wear appropriate protective equipment, including respiratory protection, as conditions warrant.
- Keep all sources of ignition and hot metal surfaces away.

2-3. Checking for presence of refrigerant
- The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable atmospheres.
- Ensure that the leak detection equipment being used is suitable for use with flammable refrigerants, i.e. non sparking, adequately sealed or intrinsically safe.
- In case of leakage/spillage happened, immediately ventilate area and stay upwind and away from spill/release.
- In case of leakage/spillage happened, do notify persons downwind of the leaking/spill, isolate immediate hazard area and keep unauthorized personnel out.

2-4. Presence of fire extinguisher
- If any hot work is to be conducted on the refrigeration equipment or any associated parts, appropriate fire extinguishing equipment shall be available at hand.
- Have a dry powder or CO_2 fire extinguisher adjacent to the charging area.

2-5. No ignition sources
- No person carrying out work in relation to a refrigeration system which involves exposing any pipe work that contains or has contained flammable refrigerant shall use any sources of ignition in such a manner that it may lead to the risk of fire or explosion. He/She must not be smoking when carrying out such work.
- All possible ignition sources, including cigarette smoking, should be kept sufficiently far away from the site of installation, repairing, removing and disposal, during which flammable refrigerant can possibly be released to the surrounding space.
- Prior to work taking place, the area around the equipment is to be surveyed to make sure that there are no flammable hazards or ignition risks.
- “No Smoking” signs shall be displayed.

2-6. Ventilated area
- Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work.
- A degree of ventilation shall continue during the period that the work is carried out.
- The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

2-7. Checks to the refrigeration equipment
- Where electrical components are being changed, they shall be fit for the purpose and to the correct specification.
- At all times the manufacturer’s maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer’s technical department for assistance.
- The following checks shall be applied to installations using flammable refrigerants.
  - The ventilation machinery and outlets are operating adequately and are not obstructed.
  - Marking to the equipment continues to be visible and legible. Markings and signs that are illegible shall be corrected.

2-8. Checks to electrical devices
- Repair and maintenance to electrical components shall include initial safety checks and component inspection procedures.
- Initial safety checks shall include but not limit to:-
  - That capacitors are discharged: this shall be done in a safe manner to avoid possibility of sparking.
  - That there is no live electrical components and wiring are exposed while charging, recovering or purging the system.
  - That there is continuity of earth bonding.
- At all times the manufacturer’s maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer’s technical department for assistance.
- If a fault exists that could compromise safety, then no electrical supply shall be connected to the circuit until it is satisfactorily dealt with.
- If the fault cannot be corrected immediately but it is necessary to continue operation, an adequate temporary solution shall be used.
- The owner of the equipment must be informed or reported so all parties are advised thereinafter.

⚠ CAUTION

3. Repairs to sealed components

- During repairs to sealed components, all electrical supplies shall be disconnected from the equipment being worked upon prior to any removal of sealed covers, etc.
- If it is absolutely necessary to have an electrical supply to equipment during servicing, then a permanently operating form of leak detection shall be located at the most critical point to warn of a potentially hazardous situation.
- Particular attention shall be paid to the following to ensure that by working on electrical components, the casing is not altered in such a way that the level of protection is affected. This shall include damage to cables, excessive number of connections, terminals not made to original specification, damage to seals, incorrect fitting of glands, etc.
- Ensure that apparatus is mounted securely.
- Ensure that seals or sealing materials have not degraded such that they no longer serve the purpose of preventing the ingress of flammable atmospheres.
- Replacement parts shall be in accordance with the manufacturer's specifications.

NOTE:	The use of silicon sealant may inhibit the effectiveness of some types of leak detection equipment. Intrinsically safe components do not have to be isolated prior to working on them.

4. Repair to intrinsically safe components

- Do not apply any permanent inductive or capacitance loads to the circuit without ensuring that this will not exceed the permissible voltage and current permitted for the equipment in use.
- Intrinsically safe components are the only types that can be worked on while live in the presence of a flammable atmosphere.
- The test apparatus shall be at the correct rating.
- Replace components only with parts specified by the manufacturer. Unspecified parts by manufacturer may result ignition of refrigerant in the atmosphere from a leak.

5. Cabling

- Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects.
- The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

6. Detection of flammable refrigerants

- Under no circumstances shall potential sources of ignition be used in the searching or detection of refrigerant leaks.
- A halide torch (or any other detector using a naked flame) shall not be used.

7. The following leak detection methods are deemed acceptable for all refrigerant systems.

- No leaks shall be detected when using detection equipment with a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.15MPa). For example, a universal sniffer.
- Electronic leak detectors may be used to detect flammable refrigerants, but the sensitivity may not be adequate, or may need recalibration.
  (Detection equipment shall be calibrated in a refrigerant-free area.)
- Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used.
- Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed and the appropriate percentage of gas (25 % maximum) is confirmed.
- Leak detection fluids are suitable for use with most refrigerants but the use of detergents containing chlorine shall be avoided as the chlorine may react with the refrigerant and corrode the copper pipe-work.
- If a leak is suspected, all naked flames shall be removed/extinguished.
- If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak. The precautions in #8 must be followed to remove the refrigerant.

8. Removal and evacuation

- When breaking into the refrigerant circuit to make repairs – or for any other purpose – conventional procedures shall be used. However, it is important that best practice is followed since flammability is a consideration.
  The following procedure shall be adhered to:

> • remove refrigerant -> • purge the circuit with inert gas -> • evacuate -> • purge again with inert gas ->
> • open the circuit by cutting or brazing

- The refrigerant charge shall be recovered into the correct recovery cylinders.
- The system shall be purged with OFN to render the appliances safe. (remark: OFN = oxygen free nitrogen, type of inert gas)
- This process may need to be repeated several times.
- Compressed air or oxygen shall not be used for this task.
- Flushing shall be achieved by breaking the vacuum in the system with OFN and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum.
- This process shall be repeated until no refrigerant is within the system.
- When the final OFN charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.
- This operation is absolutely vital if brazing operations on the pipe work are to take place.
- Ensure that the outlet for the vacuum pump is not close to any ignition sources and there is ventilation available.

9. Charging procedures

- In addition to conventional charging procedures, the following requirements shall be followed.
  - Ensure that contamination of different refrigerants does not occur when using charging equipment.
  - Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.
  - Cylinders shall be kept in an appropriate position according to the instructions.
  - Ensure that the refrigeration system is earthed prior to charging the system with refrigerant.
  - Label the system when charging is complete (if not already).
  - Extreme care shall be taken not to over fill the refrigeration system.
- Prior to recharging the system it shall be pressure tested with OFN (refer to #7).
- The system shall be leak tested on completion of charging but prior to commissioning.
- A follow up leak test shall be carried out prior to leaving the site.
- Electrostatic charge may accumulate and create a hazardous condition when charging and discharging the refrigerant.
  To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

⚠ CAUTION

10.

Decommissioning

- Before carrying out this procedure, it is essential that the technician is completely familiar with the equipment and all its details.
- It is recommended good practice that all refrigerants are recovered safely.
- Prior to the task being carried out, an oil and refrigerant sample shall be taken in case analysis is required prior to re-use of reclaimed refrigerant.
- It is essential that electrical power is available before the task is commenced.
  a) Become familiar with the equipment and its operation.
  b) Isolate system electrically.
  c) Before attempting the procedure ensure that:

  • mechanical handling equipment is available, if required, for handling refrigerant cylinders;
  • all personal protective equipment is available and being used correctly;
  • the recovery process is supervised at all times by a competent person;
  • recovery equipment and cylinders conform to the appropriate standards.

  d) Pump down refrigerant system, if possible.
  e) If a vacuum is not possible, make a manifold so that refrigerant can be removed from various parts of the system.
  f) Make sure that cylinder is situated on the scales before recovery takes place.
  g) Start the recovery machine and operate in accordance with manufacturer's instructions.
  h) Do not over fill cylinders. (No more than 80 % volume liquid charge).
  i) Do not exceed the maximum working pressure of the cylinder, even temporarily.
  j) When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed off.
  k) Recovered refrigerant shall not be charged into another refrigeration system unless it has been cleaned and checked.
- Electrostatic charge may accumulate and create a hazardous condition when charging or discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

11.

Labelling

- Equipment shall be labelled stating that it has been de-commissioned and emptied of refrigerant.
- The label shall be dated and signed.
- Ensure that there are labels on the equipment stating the equipment contains flammable refrigerant.

12.

Recovery

- When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.
- When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed.
- Ensure that the correct number of cylinders for holding the total system charge are available.
- All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i.e. special cylinders for the recovery of refrigerant).
- Cylinders shall be complete with pressure relief valve and associated shut-off valves in good working order.
- Recovery cylinders are evacuated and, if possible, cooled before recovery occurs.
- The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of flammable refrigerants.
- In addition, a set of calibrated weighing scales shall be available and in good working order.
- Hoses shall be complete with leak-free disconnect couplings and in good condition.
- Before using the recovery machine, check that it is in satisfactory working order, has been properly maintained and that any associated electrical components are sealed to prevent ignition in the event of a refrigerant release. Consult manufacturer if in doubt.
- The recovered refrigerant shall be returned to the refrigerant supplier in the correct recovery cylinder, and the relevant Waste Transfer Note arranged.
- Do not mix refrigerants in recovery units and especially not in cylinders.
- If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant.
- The evacuation process shall be carried out prior to returning the compressor to the suppliers.
- Only electric heating to the compressor body shall be employed to accelerate this process.
- When oil is drained from a system, it shall be carried out safely.

3. Specifications

3.1 WH-MDC05J3E5

Item		Unit	Refrigerant System		
Performance Test Condition			EN 14511 / EN14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	5.00		
		BTU/h	17100		
		kcal/h	4300		
Cooling EER		W/W	3.31		
		kcal/hW	2.85		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	5.00	5.00	
		BTU/h	17100	17100	
		kcal/h	4300	4300	
Heating COP		W/W	5.08	3.57	
		kcal/hW	4.37	3.07	
Heating ErP	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	5.0	5.0	6.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	6.00 / 237	5.12 / 202	4.08 / 160
	Annual Consumption	kWh	1113	2018	3625
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	4.0	5.0	4.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	4.20 / 165	3.63 / 142	2.95 / 115
	Annual Consumption	kWh	1274	2849	3338
	Class		A+++	A++	A+
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 49	Heating: 48	
		Power Level dB	Cooling: 65	Heating: 64 Heating: 59***	
Air Flow		m^3/min (ft^3/min)	Cooling: 51.0 (1800) Heating: 39.8 (1410)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (900)		
Refrigerant (R32)		kg (oz)	1.30 (45.9)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged / Maximum)		0.878 / -		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	1.70		

Item		Unit	Refrigerant System
Fan	Type		Propeller Fan
	Material		PP
	Motor Type		DC Motor (8-poles)
	Input Power	W	-
	Output Power	W	60
	Fan Speed	rpm	Cooling: 650 Heating: 520
Heat Exchanger	Fin material		Aluminium (Pre Coat)
	Fin Type		Corrugated Fin
	Row × Stage × FPI		2 × 30 × 17
	Size (W × H × L)	mm	38.1 × 762 × 903.8:873.8

Item		Unit	Mono Bloc Unit		
Power Source (Phase, Voltage, Cycle)		ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 1.51	Heating: 0.985	Heating: 1.40
Maximum Input Power For Heatpump System		kW	2.59		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 12.0 / 2.59k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	4.7		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 7.0	Heating: 4.7	Heating: 6.6
Maximum Current For Heatpump System		A	12.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		%	Cooling: 94	Heating: 91	Heating: 93
Dimension	Height	mm (inch)	865 (34-1/16)		
	Width	mm (inch)	1283 (50-17/32)		
	Depth	mm (inch)	320 (12-5/8)		
Net Weight		kg (lbs)	99 (218)		
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item			Unit	Water System
Performance Test Condition				EN 14511 / EN14825
Operation Range	Outdoor Ambient		°C	Cooling: 10 ~ 43 Heating: -20 ~ 35
	Water Outlet		°C	Cooling: 5 ~ 20 Heating: 20 ~ 55 (Below Ambient -15°C) 20 ~ 60 (Above Ambient -10°C)
Internal Pressure Differential			kPa	Cooling: 12.0 Heating: 12.0
Water Pipe Connector (in.)				(1-1/4)
Water Drain Hose Inner Diameter			mm (inch)	15.0 (19/32)
Pump	Motor Type			DC Motor
	Input Power		W	173
Hot Water Coil	Type			Brazed Plate
	No. of Plates			36
	Size (H × W × L)		mm	68.3 × 121 × 333
	Water Flow Rate		l/min (m^3/h)	Cooling: 14.3 (0.9) Heating: 14.3 (0.9)
Pressure Relief Valve Water Circuit			kPa	Open: 300, Close: 210 and below
Flow Switch				Electronic Sensor
Protection Device			A	Residual Current Circuit Breaker (40)
Expansion Vessel		Volume	l	6
		MWP	bar	3
Capacity of Integrated Electric Heater			kW	3.00

Note:

- Cooling capacities are based on outdoor air temperature of 35°C Dry Bulb with controlled indoor water inlet temperature of 12°C and water outlet temperature of 7°C.
- Heating capacities are based on outdoor air temperature of 7°C Dry Bulb (44.6°F Dry Bulb), 6°C Wet Bulb (42.8°F Wet Bulb) with controlled indoor water inlet temperature of 30°C and water outlet temperature of 35°C.
- Specification are subjected to change without prior notice for further improvement.
- Flow rate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and =5°C

*** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.

3.2 WH-MDC07J3E5

Item		Unit	Refrigerant System		
Performance Test Condition			EN 14511 / EN14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	7.00		
		BTU/h	23900		
		kcal/h	6020		
Cooling EER		W/W	3.06		
		kcal/hW	2.63		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	7.00	7.00	
		BTU/h	23900	23900	
		kcal/h	6020	6020	
Heating COP		W/W	4.76	3.40	
		kcal/hW	4.10	2.92	
Heating ErP	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	7.0	6.0	7.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	5.75 / 227	4.90 / 193	4.18 / 164
	Annual Consumption	kWh	1627	2532	4132
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	6.0	7.0	6.0
	Tbivalent / TOL	°C	2 / 2	-7 / -10	-15 /-22
	SCOP / ns	(W/W) / %	4.07 / 160	3.32 / 130	2.98 / 116
	Annual Consumption	kWh	1971	4354	4967
	Class		A+++	A++	A+
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 50	Heating: 50	
		Power Level dB	Cooling: 67	Heating: 68 Heating: 59***	
Air Flow		m^3/min (ft^3/min)	Cooling: 52.8 (1860) Heating: 43.3 (1530)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (900)		
Refrigerant (R32)		kg (oz)	1.30 (45.9)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged / Maximum)		0.878 / -		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	1.70		

Item		Unit	Refrigerant System
Fan	Type		Propeller Fan
	Material		PP
	Motor Type		DC Motor (8-poles)
	Input Power	W	-
	Output Power	W	60
	Fan Speed	rpm	Cooling: 670 Heating: 560
Heat Exchanger	Fin material		Aluminium (Pre Coat)
	Fin Type		Corrugated Fin
	Row × Stage × FPI		2 × 30 × 17
	Size (W × H × L)	mm	38.1 × 762 × 903.8:873.8

Item		Unit	Mono Bloc Unit		
Power Source (Phase, Voltage, Cycle)		ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.29	Heating: 1.47	Heating: 2.06
Maximum Input Power For Heatpump System		kW	3.71		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 17.0 / 3.71k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	7.0		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 10.5	Heating: 7.0	Heating: 9.5
Maximum Current For Heatpump System		A	17.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		%	Cooling: 95	Heating: 92	Heating: 94
Dimension	Height	mm (inch)	865 (34-1/16)		
	Width	mm (inch)	1283 (50-17/32)		
	Depth	mm (inch)	320 (12-5/8)		
Net Weight		kg (lbs)	104 (229)		
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item			Unit	Water System
Performance Test Condition				EN 14511 / EN14825
Operation Range	Outdoor Ambient		°C	Cooling: 10 ~ 43 Heating: -20 ~ 35
	Water Outlet		°C	Cooling: 5 ~ 20 Heating: 20 ~ 55 (Below Ambient -15°C) 20 ~ 60 (Above Ambient -10°C)
Internal Pressure Differential			kPa	Cooling: 22.0 Heating: 22.0
Water Pipe Connector (in.)				(1-1/4)
Water Drain Hose Inner Diameter			mm (inch)	15.0 (19/32)
Pump	Motor Type			DC Motor
	Input Power		W	173
Hot Water Coil	Type			Brazed Plate
	No. of Plates			36
	Size (H × W × L)		mm	68.3 × 121 × 333
	Water Flow Rate		l/min (m^3/h)	Cooling: 20.1 (1.2) Heating: 20.1 (1.2)
Pressure Relief Valve Water Circuit			kPa	Open: 300, Close: 210 and below
Flow Switch				Electronic Sensor
Protection Device			A	Residual Current Circuit Breaker (40)
Expansion Vessel		Volume	l	6
		MWP	bar	3
Capacity of Integrated Electric Heater			kW	3.00

Note:

- Cooling capacities are based on outdoor air temperature of 35°C Dry Bulb with controlled indoor water inlet temperature of 12°C and water outlet temperature of 7°C.
- Heating capacities are based on outdoor air temperature of 7°C Dry Bulb (44.6°F Dry Bulb), 6°C Wet Bulb (42.8°F Wet Bulb) with controlled indoor water inlet temperature of 30°C and water outlet temperature of 35°C.
- Specification are subjected to change without prior notice for further improvement.
- Flow rate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and =5°C

*** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.

3.3 WH-MDC09J3E5

Item		Unit	Refrigerant System		
Performance Test Condition			EN 14511 / EN14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	9.00		
		BTU/h	30700		
		kcal/h	7740		
Cooling EER		W/W	2.71		
		kcal/hW	2.33		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	9.00	7.45	
		BTU/h	30700	25400	
		kcal/h	7740	6410	
Heating COP		W/W	4.48	3.13	
		kcal/hW	3.85	2.69	
Heating ErP	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	7.0	7.0	7.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	5.75 / 227	4.90 / 193	4.18 / 164
	Annual Consumption	kWh	1627	2949	4132
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	6.0	8.0	6.0
	Tbivalent / TOL	°C	2 / 2	-7 / -10	-15 / -22
	SCOP / ns	(W/W) / %	4.07 / 160	3.32 / 130	2.98 / 116
	Annual Consumption	kWh	1971	4971	4967
	Class		A+++	A++	A+
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 51	Heating: 51	
		Power Level dB	Cooling: 68	Heating: 69 Heating: 59***	
Air Flow		m^3/min (ft^3/min)	Cooling: 55.4 (1960) Heating: 53.2 (1880)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	FW50S (900)		
Refrigerant (R32)		kg (oz)	1.30 (45.9)		
F-GAS	GWP		675		
	CO_2eq (ton) (Precharged / Maximum)		0.878 / -		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	1.70		

Item		Unit	Refrigerant System
Fan	Type		Propeller Fan
	Material		PP
	Motor Type		DC Motor (8-poles)
	Input Power	W	-
	Output Power	W	60
	Fan Speed	rpm	Cooling: 700 Heating: 670
Heat Exchanger	Fin material		Aluminium (Pre Coat)
	Fin Type		Corrugated Fin
	Row × Stage × FPI		2 × 30 × 17
	Size (W × H × L)	mm	38.1 × 762 × 903.8:873.8

Item		Unit	Mono Bloc Unit		
Power Source (Phase, Voltage, Cycle)		ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 3.32	Heating: 2.01	Heating: 2.38
Maximum Input Power For Heatpump System		kW	3.83		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 17.0 / 3.83k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	9.3		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 14.7	Heating: 9.3	Heating: 11.0
Maximum Current For Heatpump System		A	17.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		%	Cooling: 98	Heating: 94	Heating: 94
Dimension	Height	mm (inch)	865 (34-1/16)		
	Width	mm (inch)	1283 (50-17/32)		
	Depth	mm (inch)	320 (12-5/8)		
Net Weight		kg (lbs)	104 (229)		
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item			Unit	Water System
Performance Test Condition				EN 14511 / EN14825
Operation Range	Outdoor Ambient		°C	Cooling: 10 ~ 43 Heating: -20 ~ 35
	Water Outlet		°C	Cooling: 5 ~ 20 Heating: 20 ~ 55 (Below Ambient -15°C) 20 ~ 60 (Above Ambient -10°C)
Internal Pressure Differential			kPa	Cooling: 32.0 Heating: 32.0
Water Pipe Connector (in.)				(1-1/4)
Water Drain Hose Inner Diameter			mm (inch)	15.0 (19/32)
Pump	Motor Type			DC Motor
	Input Power		W	173
Hot Water Coil	Type			Brazed Plate
	No. of Plates			36
	Size (W × H × L)		mm	68.3 × 121 × 333
	Water Flow Rate		l/min (m^3/h)	Cooling: 25.8 (1.5) Heating: 25.8 (1.5)
Pressure Relief Valve Water Circuit			kPa	Open: 300, Close: 210 and below
Flow Switch				Electronic Sensor
Protection Device			A	Residual Current Circuit Breaker (40)
Expansion Vessel		Volume	l	6
		MWP	bar	3
Capacity of Integrated Electric Heater			kW	3.00

Note:

- Cooling capacities are based on outdoor air temperature of 35°C Dry Bulb with controlled indoor water inlet temperature of 12°C and water outlet temperature of 7°C.
- Heating capacities are based on outdoor air temperature of 7°C Dry Bulb (44.6°F Dry Bulb), 6°C Wet Bulb (42.8°F Wet Bulb) with controlled indoor water inlet temperature of 30°C and water outlet temperature of 35°C.
- Specification are subjected to change without prior notice for further improvement.
- Flow rate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and =5°C

*** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.

4. Features

- **Inverter Technology**
  - Energy saving
- **High Efficiency**
- **Compact Design**
- **Environment Protection**
  - Non-ozone depletion substances refrigerant (R410A)
- **Easy to use control panel**
- **Weekly Timer setting**
- **Quality Improvement**
  - Random auto restart after power failure for safety restart operation
  - Gas leakage protection
  - Prevent compressor reverse cycle
  - Inner protector to protect compressor
- **Serviceability Improvement**
  - Breakdown Self Diagnosis function
  - System Status Check Buttons for servicing purpose
  - System Pumpdown Button for servicing purpose
  - Front maintenance design for outdoor unit
- **Operation Condition**

	HEATING (CIRCUIT)	*1, *2 COOLING (CIRCUIT)
Water outlet temperature (°C) (Min. / Max.)	20 / 55 (Below Ambient -15 °C) *3 20 / 60 (Above Ambient -10 °C) *3	5 / 20
Outdoor ambient temperature (°C) (Min. / Max.)	-20 / 35	10 / 43

NOTICE: When the outdoor temperature is out of the range in the table, the heating capacity will drop significantly and the mono bloc unit may stop operating for its protection.
The unit will restart automatically after the outdoor temperature returns to the specified range.

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorized service partners.

*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

*3 Between outdoor ambient -10°C and -15°C, the water outlet temperature gradually decreases from 60°C to 55°C.

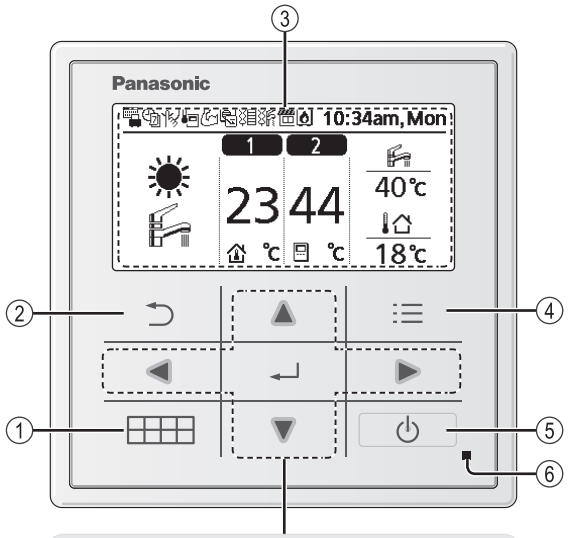
5. Location of Controls and Components

5.1 Indoor Unit

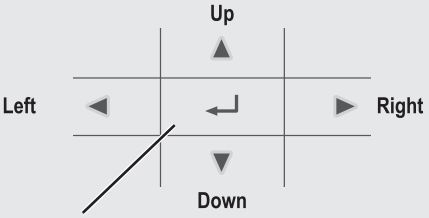
5.1.1 Location of Control

Buttons / Indicator

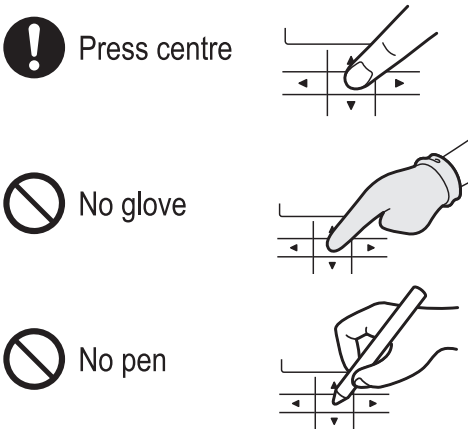
Buttons / Indicator	
①	**Quick Menu button** (For more details, refer to the separate Quick Menu Guide.)
②	**Back button** Returns to the previous screen
③	**LCD Display**
④	**Main Menu button** For function setup
⑤	**ON/OFF button** Starts/Stops operation
⑥	**Operation indicator** Illuminates during operation, blinks during alarm.

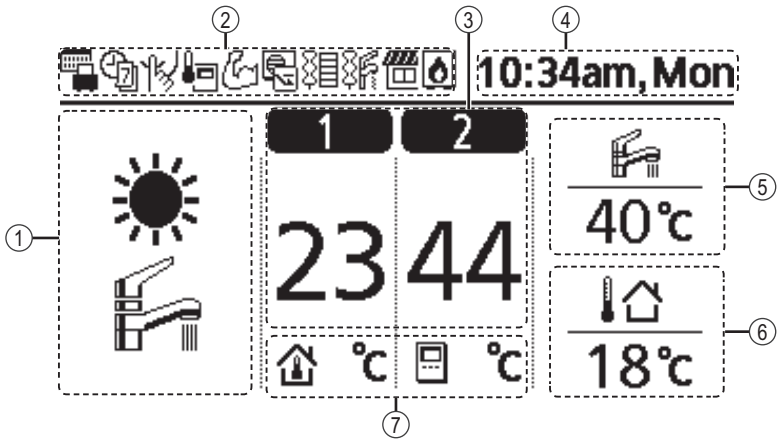


Cross key buttons
Selects an item.



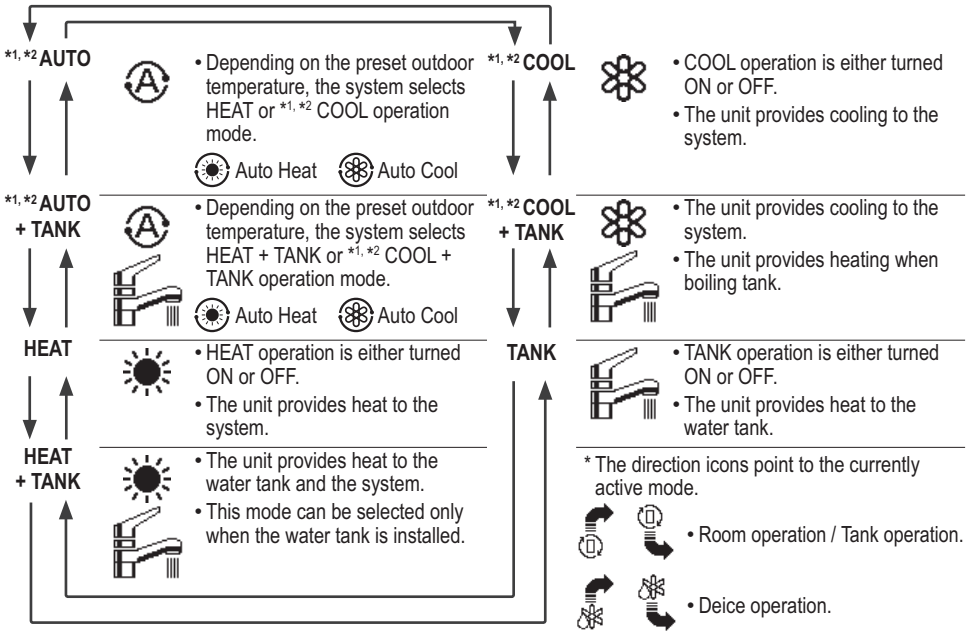
Enter button
Fixes the selected content.





Display

① **Mode selection**



② **Operation icons**

The status of operation is displayed.
Icon will not display (under operation OFF screen) whenever operation is OFF except weekly timer.

- Holiday operation status
- Weekly Timer operation status
- Quiet operation status
- Zone:Room Thermostat →Internal sensor status
- Powerful operation status
- Demand Control or SG ready or SHP status
- Room Heater status
- Tank Heater status
- Solar status
- Bivalent status (Boiler)

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

③ **Temperature of each zone**

④ **Time and day**

⑤ **Water Tank temperature**

⑥ **Outdoor temperature**

⑦ **Sensor type/Set temperature type icons**

- Water Temperature →Compensation curve
- Water Temperature →Direct
- Pool only
- Room Thermostat →External
- Room Thermostat →Internal
- Room Thermistor

5.1.2 Initialization

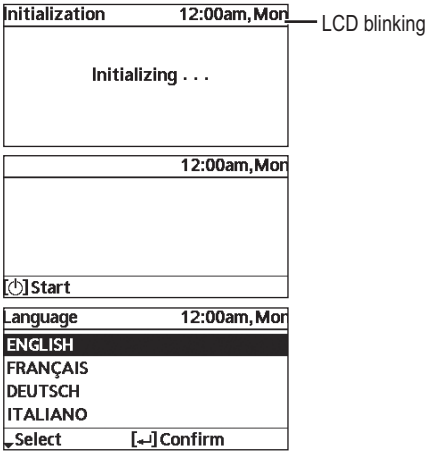
Before starting to install the various menu settings, please initiate the Remote Controller by selecting the language of operation and installing the date and time correctly.
When power is turned on for the first time, it becomes the setting screen automatically. It can also be set from personal setting of the menu.

Selecting the language

Wait while the display is initializing.
When initializing screen ends, it turns to normal screen.
When any button is pressed, language setting screen appears.

① **Scroll with ▼ and ▲ to select the language.**

② **Press ↵ to confirm the selection.**



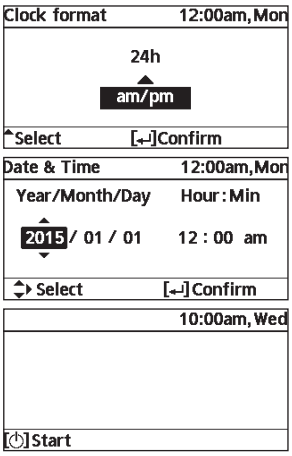
Setting the clock

① **Select with ▼ or ▲ how to display the time, either 24h or am/pm format (for example, 15:00 or 3:00 pm).**

② **Press ↵ to confirm the selection.**

③ **Use ▼ and ▲ to select year, month, day, hour and minutes. (Select and move with ▶ and press ↵ to confirm.)**

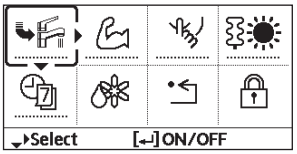
④ **Once the time is set, time and day will appear on the display even if the Remote Controller is turned OFF.**



5.1.3 Quick Menu

After the initial settings have been completed, you can select a quick menu from the following options and edit the setting.

① **Press ⊞ to display the quick menu.**



Force DHW | Powerful | Quiet | Force Heater
Weekly Timer | Force Defrost | Error Reset | R/C Lock

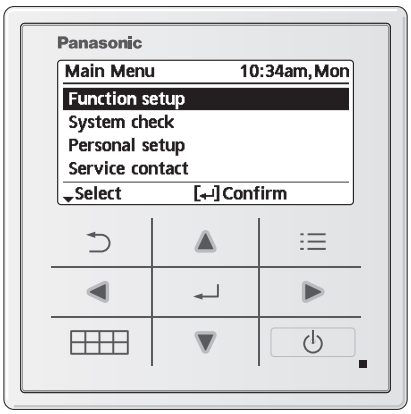
② **Use ▲ ▼ ◀ ▶ to select menu.**

③ **Press ↵ to turn on/off the select menu.**

5.1.4 Menus For user

Select menus and determine settings according to the system available in the household. All initial settings must be done by an authorised dealer or a specialist. It is recommended that all alterations of the initial settings are also done by an authorised dealer or a specialist.

- After initial installation, you may manually adjust the settings.
- The initial setting remains active until the user changes it.
- The Remote Controller can be used for multiple installations.
- Ensure the operation indicator is OFF before setting.
- The system may not work properly if set wrongly. Please consult an authorised dealer.

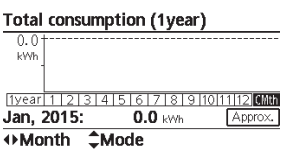
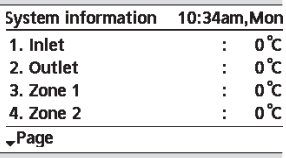
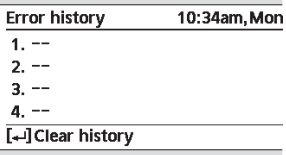
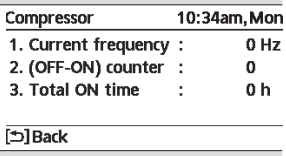
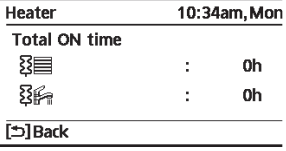
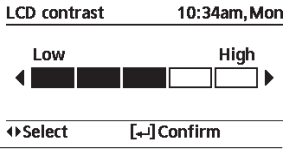


To display <Main Menu>: ☰

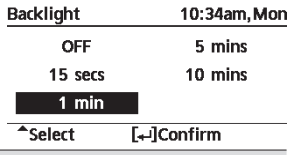
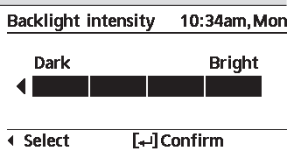
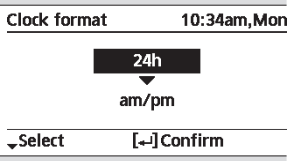
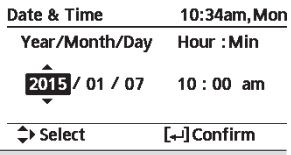
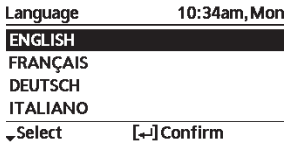
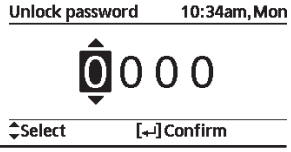
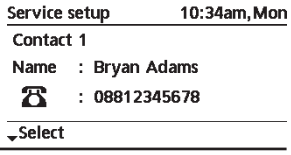
To select menu: ▲ ▼ ◀ ▶

To confirm the selected content: ↵

Menu	Default Setting	Setting Options / Display
1 Function setup		
1.1 > Weekly timer		
Once the weekly timer is set up, User can edit from Quick Menu. To set up to 6 patterns of operation on a daily basis. • Disabled if Heat-Cool SW is select "Yes" or if Force Heater is on.	**Timer setup** Select day of the week and set the patterns needed (Time / Operation ON/OFF / Mode) **Timer copy** Select day of the week	Weekly timer 10:34am,Mon Sun Mon Tue Wed Thu Fri Sat 1. 8:00am ON 40℃ 2. 12:00pm ON 24/28℃ 40℃ 3. 1:00pm ON 12/10℃ ◆▸Day ▾Pattern [↵]Edit
1.2 > Holiday timer		
To save energy, a holiday period may be set to either turn OFF the system or lower the temperature during the period.	OFF	ON OFF
	> ON	
	Holiday start and end. Date and time	Holiday: End 10:34am,Mon Year/Month/Day Hour : Min 2015 / 01 / 07 10 : 00 am ◆▸ Select [↵]Confirm
	OFF or lowered temperature	
• Weekly timer setting may be temporarily disabled during Holiday timer setting but it will be restored once the Holiday timer is completed.		
1.3 > Quiet timer		
To operate quietly during the preset period. 6 patterns may be set. Level 0 means the mode is off.	Time to start Quiet : Date and time Level of quietness: 0 ~ 3	Quiet 10:34am,Mon Pattern Time Level 1 8:00am 0 2 5:00pm 1 3 11:00pm 3 ▾Select [↵]Edit
1.4 > Room heater		
To set the room heater ON or OFF.	OFF	ON OFF
1.5 > Tank heater		
To set the tank heater ON or OFF.	OFF	ON OFF
• Available only if connected to the tank.		
1.6 > Sterilization		
To set the auto sterilization ON or OFF.	OFF	ON OFF
• Available only if connected to the tank. • Do not use the system during sterilization in order to prevent scalding with hot water, or overheating of shower. • Ask an authorised dealer to determine the level of sterilization function field settings according to the local laws and regulations.		

Menu	Default Setting	Setting Options / Display
2 System check		
2.1 > Energy monitor		
Present or historical chart of energy consumption, generation or COP.	**Present** Select and retrieve **Historical chart** Select and retrieve	
• COP= Coefficient of Performance. • For historical chart, the period is selected from 1 day/1 week/1year. • Energy consumption (kWh) of heating, *1, *2 cooling, tank and total may be retrieved. • The total power consumption is an estimated value based on AC 230 V and may differ from value measured by precise equipment.		
2.2 > System information		
Shows all system information in each area.	**Actual system information of 10 items: Inlet / Outlet / Zone 1 / Zone 2 / Tank / Buffer tank / Solar / Pool / COMP frequency / Pump flowrate** Select and retrieve	
2.3 > Error history		
• Refer to Troubleshooting for error codes. • The most recent error code is displayed at the top.	Select and retrieve	
2.4 > Compressor		
Shows the compressor performance.	Select and retrieve	
2.5 > Heater		
Total hours of ON time for Room heater/Tank heater.	Select and retrieve	
3 Personal setup		
3.1 > Touch sound		
Turns the operation sound ON/ OFF.	ON	
3.2 > LCD contrast		
Sets the screen contrast.	3	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
3.3 > Backlight		
Sets the duration of screen backlight.	1 min	
3.4 > Backlight intensity		
Sets screen backlight brightness.	4	
3.5 > Clock format		
Sets the type of clock display.	24h	
3.6 > Date & Time		
Sets the present date and time.	Year / Month / Day / Hour / Min	
3.7 > Language		
Sets the display language for the top screen. • For Greek, please refer to the English version.	ENGLISH / FRANÇAIS / DEUTSCH / ITALIANO / ESPAÑOL / DANISH / SWEDISH / NORWEGIAN / POLISH / CZECH / NEDERLANDS / TÜRKÇE / SUOMI / MAGYAR / SLOVENŠČINA / HRVATSKI	
3.8 > Unlock password		
4 digit password for all the settings.	0000	
4 Service contact		
4.1 > Contact 1 / Contact 2		
Preset contact number for installer.	Select and retrieve	

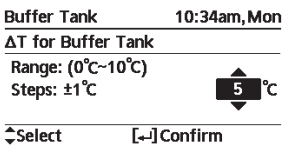
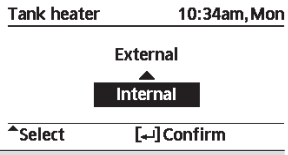
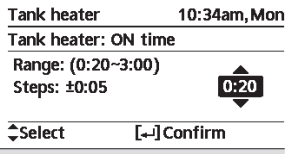
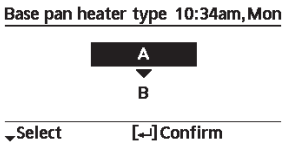
5.1.5 Menus For installer

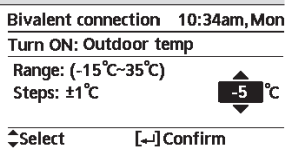
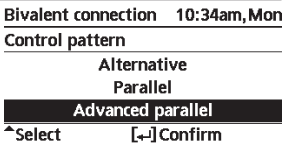
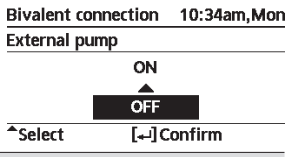
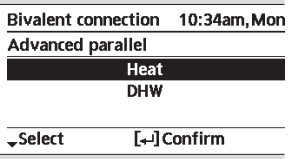
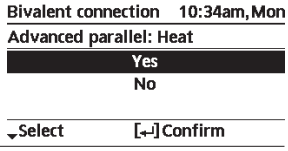
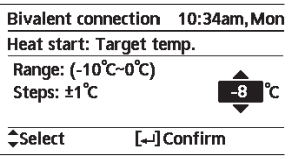
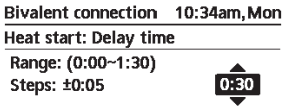
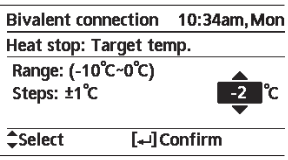
Menu	Default Setting	Setting Options / Display
5 Installer setup > System setup		
5.1 > Optional PCB connectivity		
To connect to the external PCB required for servicing.	No	Yes ▲ No

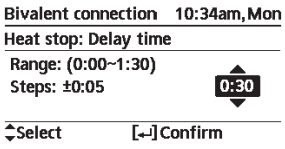
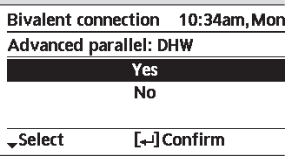
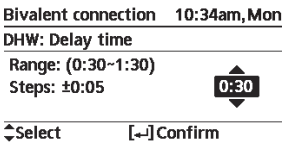
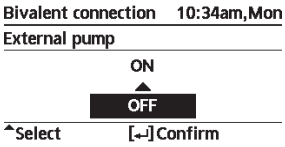
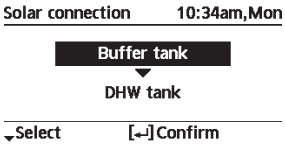
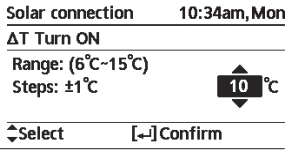
• If the external PCB is connected (optional), the system will have following additional functions:

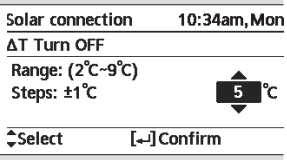
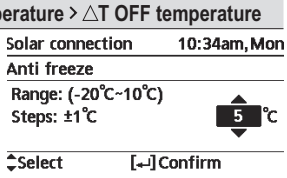
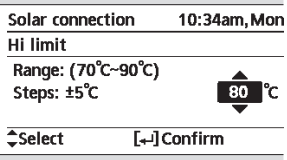
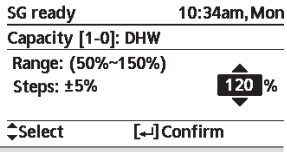
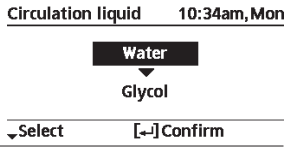
① Buffer tank connection and control over its function and temperature.
② Control over 2 zones (including the swimming pool and the function to heat water in it).
③ Solar function (the solar thermal panels connected to either the DHW (Domestic Hot Water) Tank or the Buffer Tank.
• DHW is not applicable for WH-ADC *models.
④ External compressor switch.
⑤ External error signal.
⑥ SG ready control.
⑦ Demand control.
⑧ Heat-Cool SW

5.2 > Zone & Sensor		
To select the sensors and to select either 1 zone or 2 zone system.	**Zone** • After selecting 1 or 2 zone system, proceed to the selection of room or swimming pool. • If the swimming pool is selected, the temperature must be selected for △T temperature between 0°C ~ 10 °C.	Zone & Sensor 10:34am,Mon Zone **1 Zone system** 2 Zones system ▼Select [↵]Confirm
	Sensor * For room thermostat, there is a further selection of external or internal.	Zone & Sensor 10:34am,Mon Sensor **Water temperature** Room thermostat Room thermistor ▼Select [↵]Confirm
5.3 > Heater capacity		
To reduce the heater power if unnecessary.* 3 kW / 6 kW / 9kW * Options of kW vary depending on the model.		Heater capacity 10:34am,Mon **3 kW** [↵]Confirm
5.4 > Anti freezing		
To activate or deactivate the water freeze prevention when the system is OFF	Yes	**Yes** ▼ No
5.5 > Tank connection		
To connect tank to the system.	No	Yes ▲ No
5.6 > DHW capacity		
To select tank heating capacity to variable or standard. Variable capacity heat up tank with fast mode and keep the tank temperature with efficient mode. While standard capacity heat up tank with rated heating capacity. * This option is available if Tank connection is selected (YES).	Variable	**Variable** ▼ Standard

Menu	Default Setting	Setting Options / Display
5.7 > Buffer tank connection		
To connect tank to the system and if selected YES, to set △T temperature. • The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display.	No	
	> Yes	
	5 °C	Set △T for Buffer Tank 
5.8 > Tank heater		
To select external or internal tank heater and if External is selected, set a timer for the heater to come on. * This option is available if Tank connection is selected (YES).	Internal	
	> External	
	0:20	Tank heater ON time set. 
5.9 > Base pan heater		
To select whether or not optional base pan heater is connected. * Type A - The base pan heater activates only during deice operation. * Type B - The base pan heater activates when outdoor ambient temperature is 5 °C or lower.	No	
	> Yes	
	A	Set base pan heater type*. 
5.10 > Alternative outdoor sensor		
To select an alternative outdoor sensor.	No	
5.11 > Bivalent connection		
To select to enable or disable bivalent connection.	No	
> Yes		
To select either auto control pattern or SG ready input control pattern. * This selection only display to select when optional pcb connection set to Yes.	Auto	

Menu	Default Setting	Setting Options / Display
To select a bivalent connection to allow an additional heat source such as a boiler to heat-up the buffer tank and domestic hot water tank when heatpump capacity is insufficient at low outdoor temperature. The bivalent feature can be set-up either in alternative mode (heatpump and boiler operate alternately), or in parallel mode (both heatpump and boiler operate simultaneously), or in advance parallel mode (heatpump operates and boiler turns on for buffer-tank and/or domestic hot water depending on the control pattern setting options).	**> Yes > Auto**	
	-5 °C — Set outdoor temperature for turn ON Bivalent connection.	
	Yes > After selecting the outdoor temperature	
	Control pattern Alternative / Parallel / Advanced parallel • Select advanced parallel for bivalent use of the tanks.	
	Control pattern > Alternative	
	OFF — Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.	
	Control pattern > Advanced parallel	
	Heat — Selection of the tank • "Heat" implies Buffer Tank and "DHW" implies Domestic Hot Water Tank.	
	Control pattern > Advanced parallel > Heat > Yes	
	• Buffer Tank is activated only after selecting "Yes".	
	-8 °C — Set the temperature threshold to start the bivalent heat source.	
	0:30 — Delay timer to start the bivalent heat source (in hour and minutes).	
	-2 °C — Set the temperature threshold to stop the bivalent heat source.	

Menu	Default Setting	Setting Options / Display
	0:30	Delay timer to stop the bivalent heat source (in hour and minutes). 
	Control pattern > Advanced parallel > DHW > Yes	
	• DHW Tank is activated only after selecting "Yes".	
	0:30	Delay timer to start the bivalent heat source (in hour and minutes). 
SG ready input control for bivalent system follow below input condition. SG signal: Vcc-bit1 / Vcc-bit2 Open / Open: Heat Pump OFF, Boiler OFF Short / Open: Heat Pump ON, Boiler OFF Open / Short: Heat Pump OFF, Boiler ON Short / Short: Heat Pump ON, Boiler ON	**> Yes > SG ready**	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection. 
5.12 > External SW		
	No	
5.13 > Solar connection		
• The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display. • DHW is not applicable for WH-ADC *models.	No	
	> Yes	
	Buffer tank	Selection of the tank 
	> Yes > After selecting the tank	
	10 °C	Set △T ON temperature 

Menu	Default Setting	Setting Options / Display
	> Yes > After selecting the tank > △T ON temperature	
	5 °C	Set △T OFF temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature	
	5 °C	Set Antifreeze temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature > After setting the antifreeze temperature	
	80 °C	Set Hi limit 
5.14	**> External error signal**	
	No	
5.15	**> Demand control**	
	No	
5.16	**> SG ready**	
	No	
	> Yes	
	120 %	Capacity (1) & (2) of DHW (in %), Heat (in %) and Cool (in °C) 
5.17	**> External compressor SW**	
	No	
5.18	**> Circulation liquid**	
To select whether to circulate water or glycol in the system.	Water	

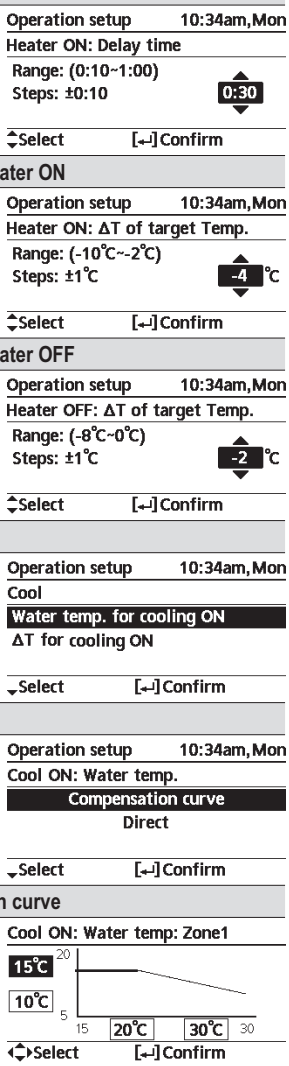
Menu	Default Setting	Setting Options / Display
5.19 > Heat-Cool SW		
	No	Yes ▲ No
5.20 > Force heater		
To turn on Force heater either manually (by default) or automatically.	Manual	Force heater 10:34am,Mon Auto ▲ Manual ▲Select [↵]Confirm
5.21 > Force defrost		
If auto selection is set, the unit will start defrost operation if long heating hour operate during low outdoor temperature.	Manual	Auto ▲ Manual
5.22 > Defrost signal		
To turn on defrost signal to stop fan coil during defrost operation. (If defrost signal set to yes, bivalent function will not available to use)	No	Yes ▲ No
5.23 > Pump flowrate		
To set variable flow pump control or fix pump duty control.	△T	ΔT ▼ Max. Duty

6 Installer setup > Operation setup		
To access to the four major functions or modes.	4 main modes Heat / *1, *2 Cool / *1, *2 Auto / Tank	Operation setup 10:34am,Mon Heat Cool Auto Tank ▼Select [↵]Confirm
6.1 > Heat		
To set various water & ambient temperatures for heating.	Water temp. for heating ON / Outdoor temp. for heating OFF / △T for heating ON / Heater ON/OFF	Operation setup 10:34am,Mon Heat Water temp. for heating ON Outdoor temp. for heating OFF ΔT for heating ON ▼Select [↵]Confirm
	> Water temp. for heating ON	
	Compensation curve Heating ON temperatures in compensation curve or direct input.	Operation setup 10:34am,Mon Heat ON: Water temp. Compensation curve Direct ▼Select [↵]Confirm

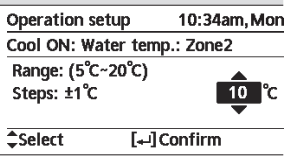
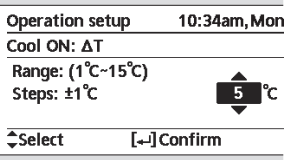
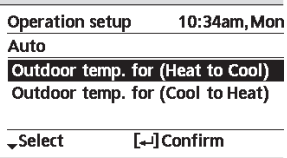
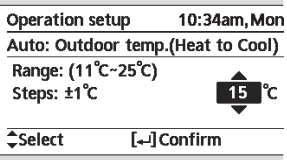
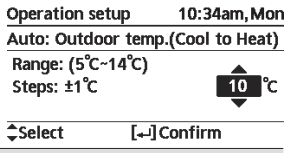
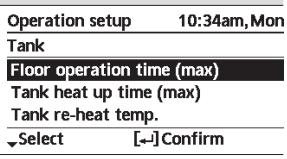
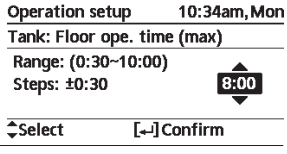
*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Water temp. for heating ON > Compensation curve	
	X axis: -5 °C, 15 °C Y axis: 55 °C, 35 °C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis).
	• Temperature range: X axis: -20 °C ~ 15 °C, Y axis: See below • Temperature range for the Y axis input: 1. WH-MDC model: 20 °C ~ 60 °C • If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Water temp. for heating ON > Direct	
	35 °C	Temperature for heating ON
	• Min. ~ Max. range is conditional as follows: 1. WH-MDC model: 20 °C ~ 60 °C • If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Outdoor temp. for heating OFF	
	24 °C	Temperature for heating OFF
	> △T for heating ON	
	5 °C	Set △T for heating ON. * This setting will not available to set when pump flowrate set to Max. duty.
	> Heater ON/OFF	
	> Heater ON/OFF > Outdoor temp. for heater ON	
	0 °C	Temperature for heater ON

Menu	Default Setting	Setting Options / Display
	> Heater ON/OFF > Delay time for heater ON	
	0:30 min	Delay time for heater to turn on
	> Heater ON/OFF > Water temperature for heater ON	
	-4 °C	Setting of water temperature to turn on from water set temperature.
	> Heater ON/OFF > Water temperature for heater OFF	
	-2 °C	Setting of water temperature to turn off from water set temperature.
6.2 **> *1, *2 Cool**		
To set various water & ambient temperatures for cooling.	Water temperatures for cooling ON and △T for cooling ON.	
	> Water temp. for cooling ON	
	Compensation curve	Cooling ON temperatures in compensation curve or direct input.
	> Water temp. for cooling ON > Compensation curve	
	X axis: 20 °C, 30 °C Y axis: 15 °C, 10 °C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis)
	• If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	

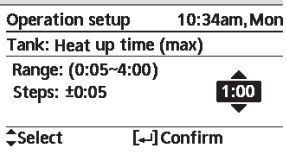


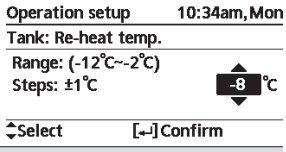
*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

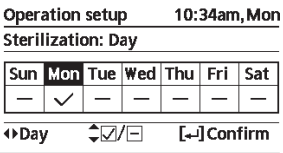
Menu	Default Setting	Setting Options / Display
	> Water temp. for cooling ON > Direct	
	10 °C	Set temperature for Cooling ON 
	• If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> △T for cooling ON	
	5 °C	Set △T for cooling ON * This setting will not available to set when pump flowrate set to Max. duty. 
6.3 **> *1, *2 Auto**		
Automatic switch from Heat to Cool or Cool to Heat.	Outdoor temperatures for switching from Heat to Cool or Cool to Heat. Outdoor temp. for (Heat to Cool) / Outdoor temp. for (Cool to Heat)	
	> Outdoor temp. for (Heat to Cool)	
	15 °C	Set outdoor temperature for switching from Heat to Cool. 
	> Outdoor temp. for (Cool to Heat)	
	10 °C	Set outdoor temperature for switching from Cool to Heat. 
6.4 **> Tank**		
Setting functions for the tank. • Available only if connected to the tank.	Floor operation time (max) / Tank heat up time (max) / Tank re-heat temp. / Sterilization	
	• The display will show 3 functions at a time.	
	> Floor operation time (max)	
	8:00	Maximum time for floor operation (in hours and minutes) 

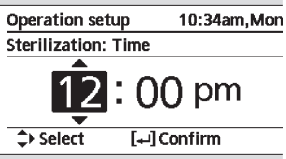
*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

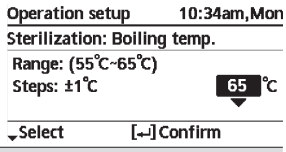
Menu	Default Setting	Setting Options / Display
	> Tank heat up time (max)	
	1:00	Maximum time for heating the tank (in hours and minutes)
	> Tank re-heat temp.	
	-8 °C	Set temperature to perform reboil of tank water.
	> Sterilization	
	Monday	Sterilization may be set for 1 or more days of the week. Sun / Mon / Tue / Wed / Thu / Fri / Sat
	> Sterilization: Time	
	12:00	Time of the selected day(s) of the week to sterilize the tank 0:00 ~ 23:59
	> Sterilization: Boiling temp.	
	65 °C	Set boiling temperatures for sterilize the tank.
	> Sterilization: Ope. time (max)	
	0:10	Set sterilizing time (in hours and minutes)

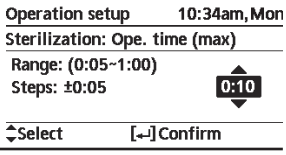












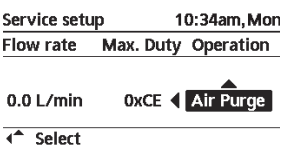
7 Installer setup > Service setup

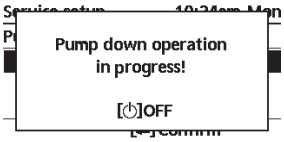
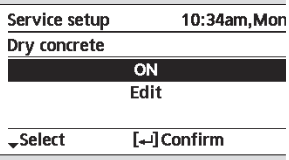
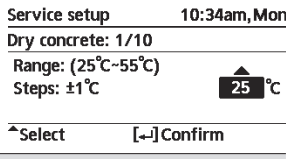
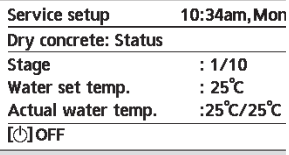
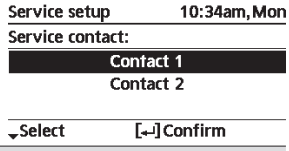
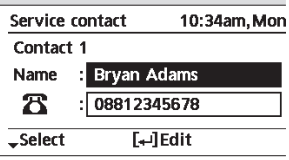
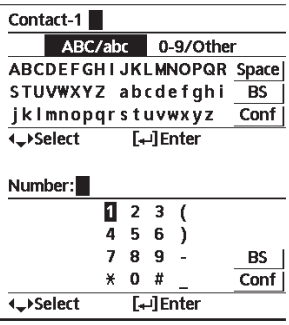
7.1 > Pump maximum speed

To set the maximum speed of the pump.

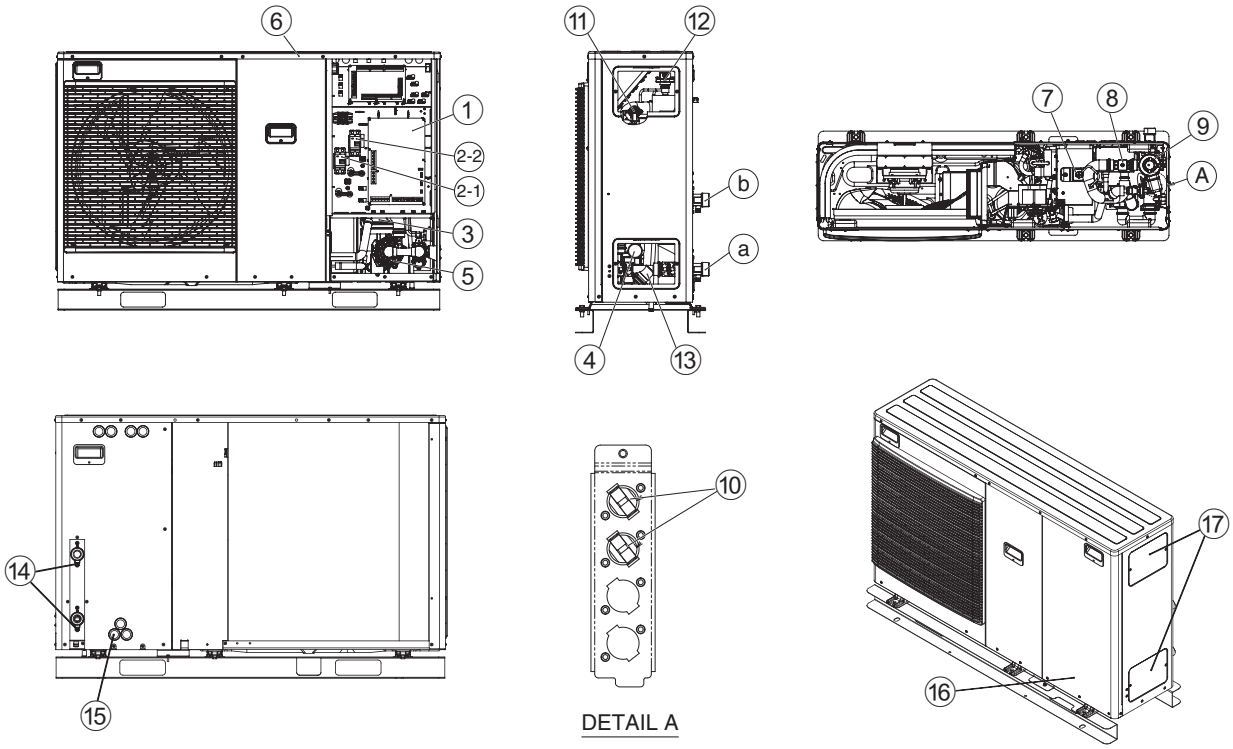
Setting the flow rate, max. duty and operation ON/OFF of the pump.

Flow rate: XX:X L/min
Max. Duty: 0x40 ~ 0xFE,
Pump: ON/OFF/Air Purge



Menu	Default Setting	Setting Options / Display
7.2 > Pump down		
To set the pump down operation.	**Pump down operation** ON	
7.3 > Dry concrete		
To dry the concrete (floor, walls, etc.) during construction. Do not use this menu for any other purposes and in period other than during construction	Edit to set the temperature of dry concrete. ON / Edit	
	> Edit	
	Stages: 1 Temperature: 25 °C Heating temperature for drying the concrete. Select the desired stages: 1 ~ 10, range: 1 ~ 99	
	> ON	
	Confirm the setting temperatures of dry concrete for each stage.	
7.4 > Service contact		
To set up to 2 contact names and numbers for the User.	Service engineer's name and contact number. Contact 1 / Contact 2	
	> Contact 1 / Contact 2	
	Contact name or number. Name / phone icon	
	Input name and number Contact name: alphabet a ~ z. Contact number: 1 ~ 9	

5.1.6 Main Components



Component name

① PCB
②-1 Single phase RCCB (Main Power)
②-2 Single phase RCCB (Backup Heater)
③ Heat Exchanger
④ Water Pressure Gauge
⑤ Water Pump
⑥ Cabinet Top Plate
⑦ Expansion Vessel
⑧ Flow Sensor
⑨ Heater Assembly
⑩ Overload Protector (2 pieces)
⑪ Pressure Relief Valve
⑫ Air Purge Valve
⑬ Magnetic Water Filter Set
⑭ Plug (2 pieces)
⑮ Bushing (7 pieces)
⑯ Cabinet Front Plate
⑰ Cover (2 pieces)

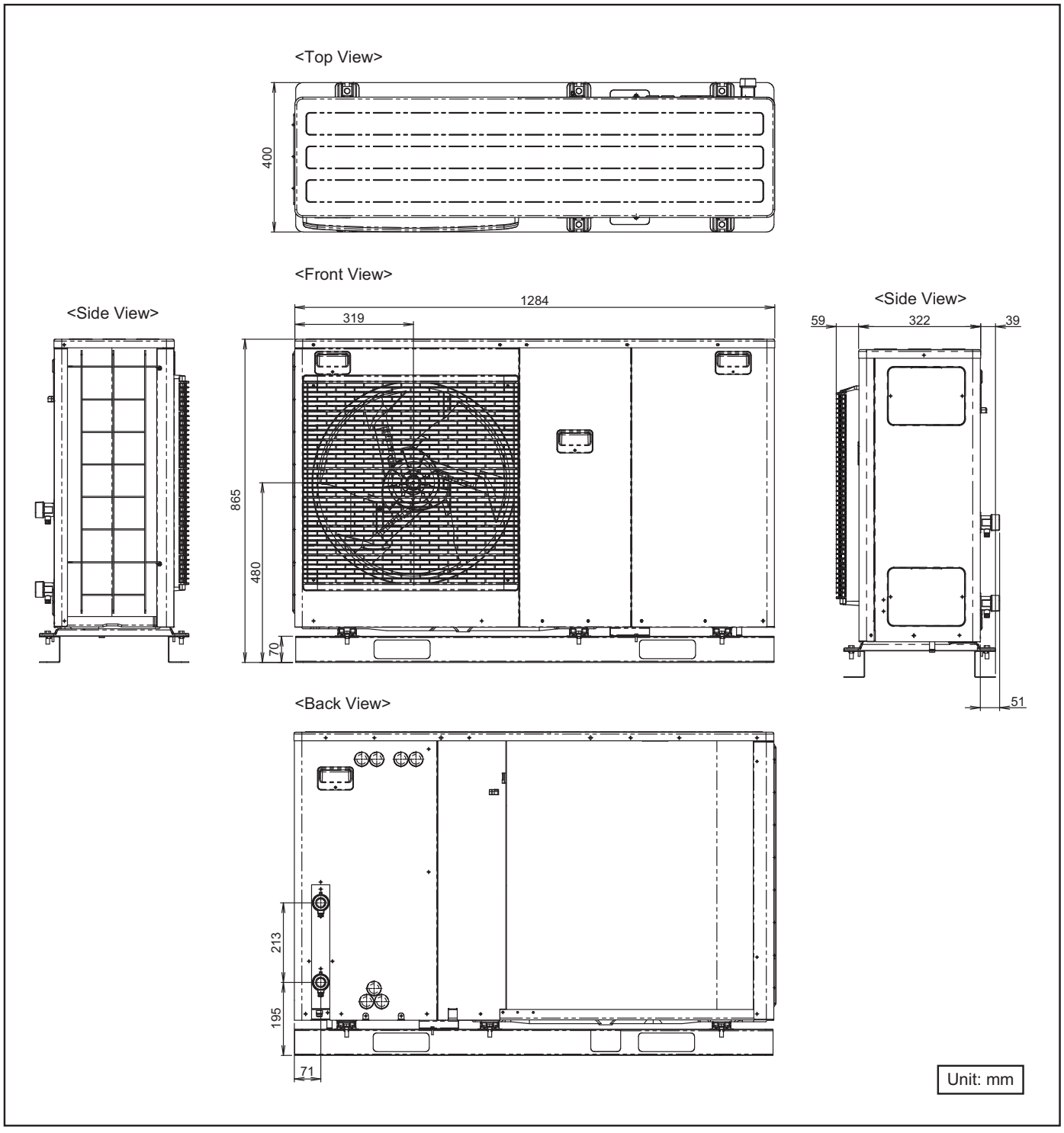
Connector name

ⓐ Water inlet
ⓑ Water outlet

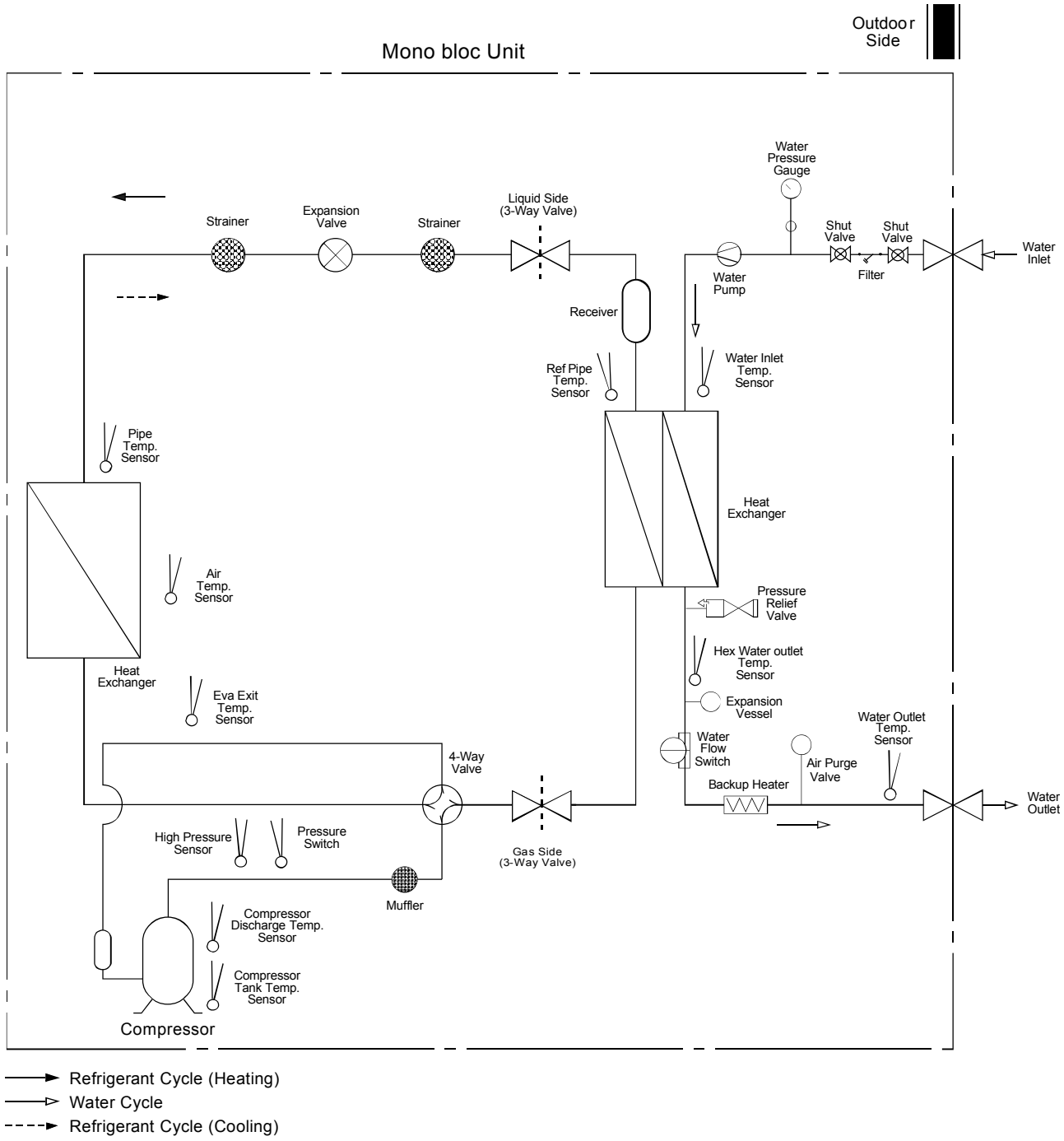
It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.

6. Dimensions

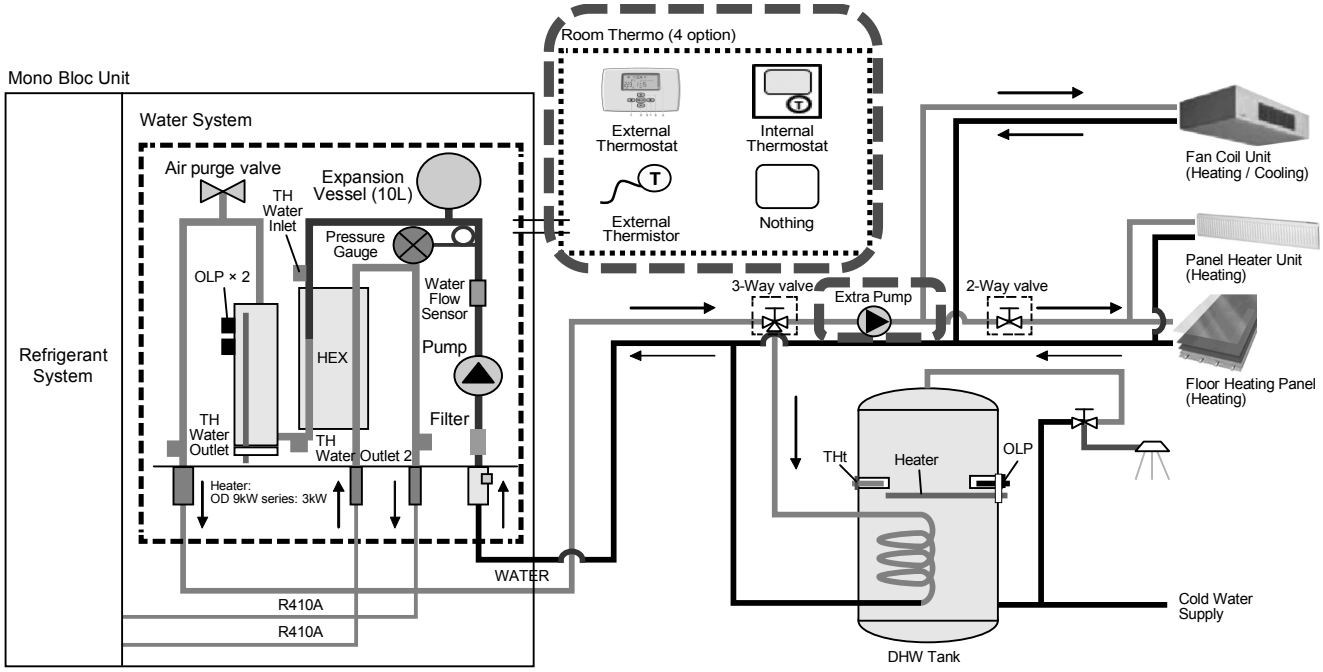
6.1 Mono Bloc Unit



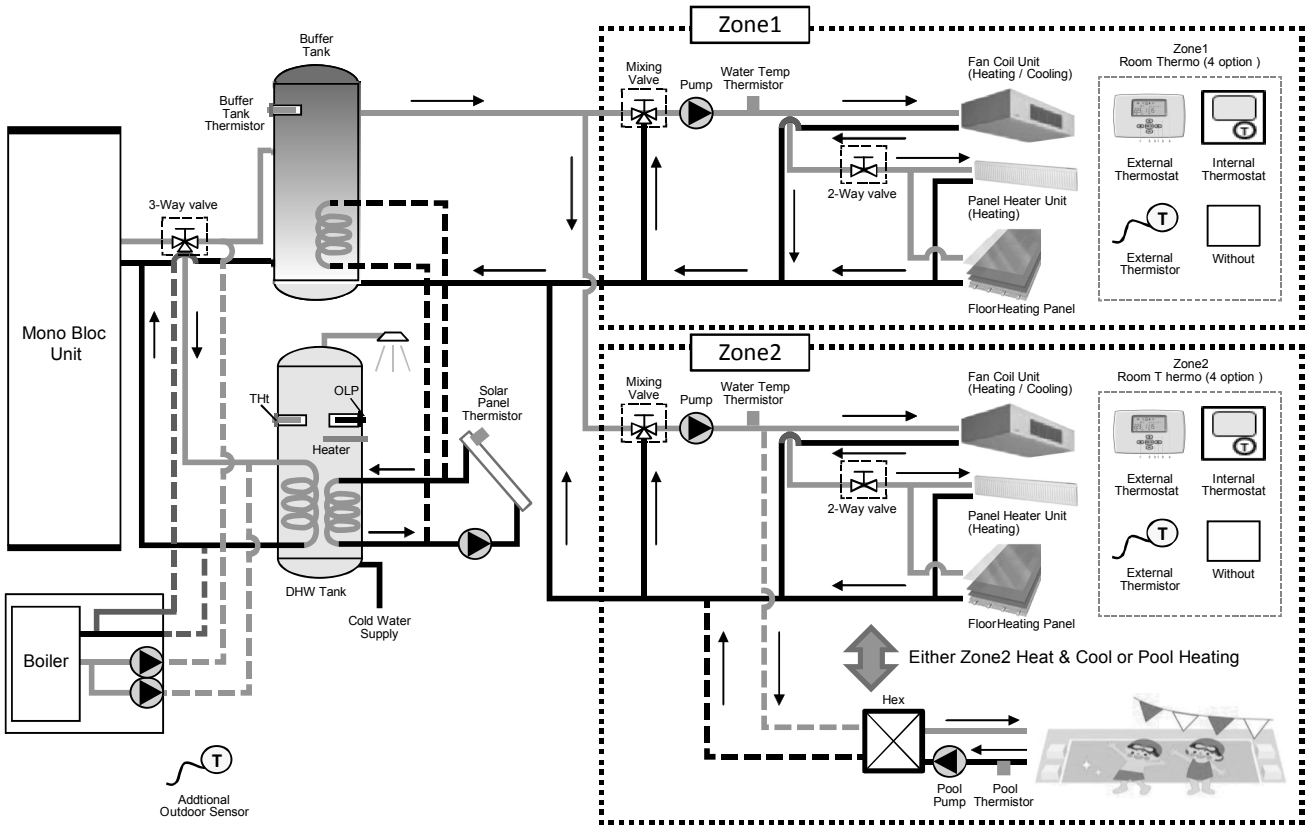
7. Refrigeration and Water Cycle Diagram



7.1 Standard System (Optional PCB not connected)

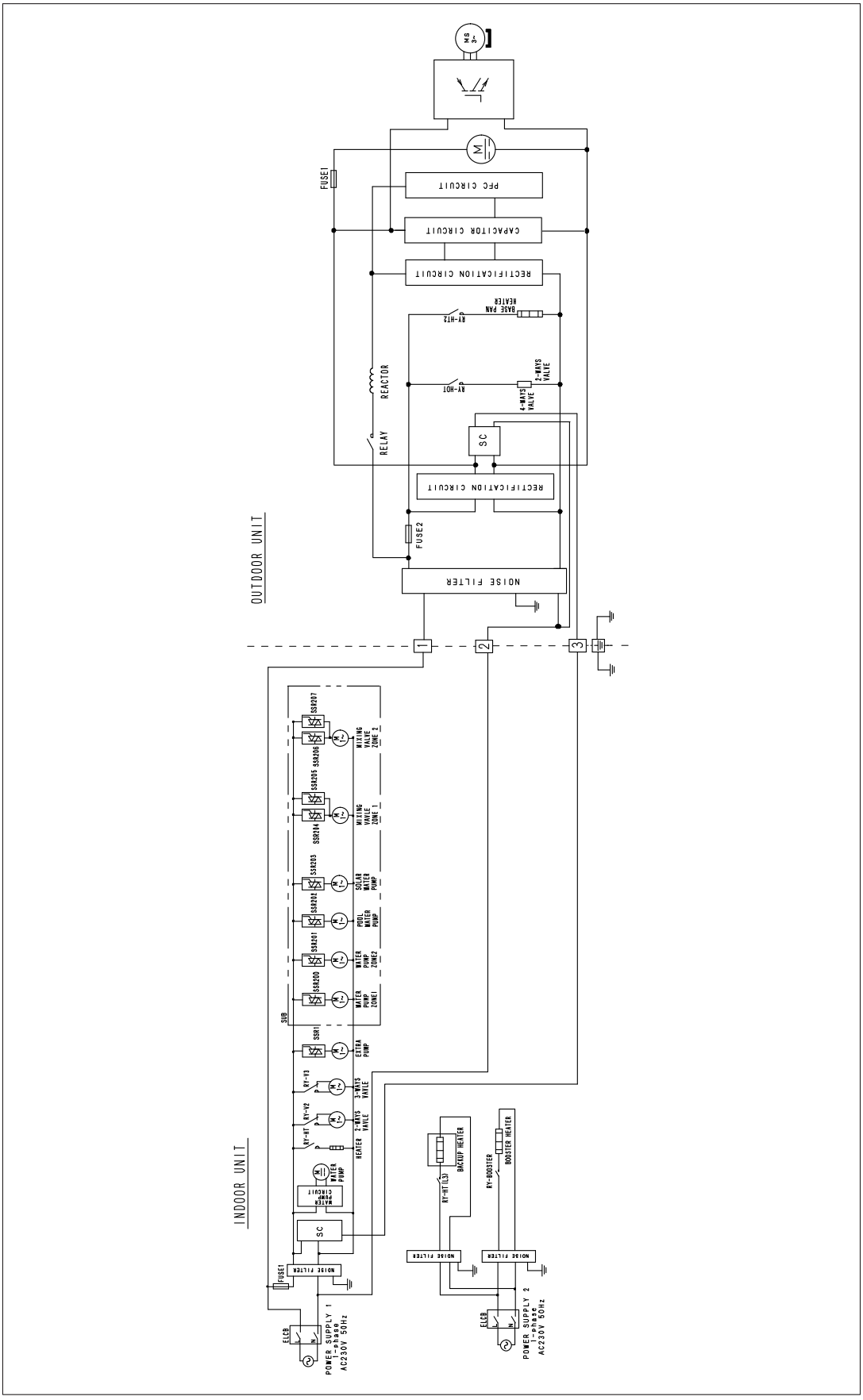


7.2 Advance System (Optional PCB connected)

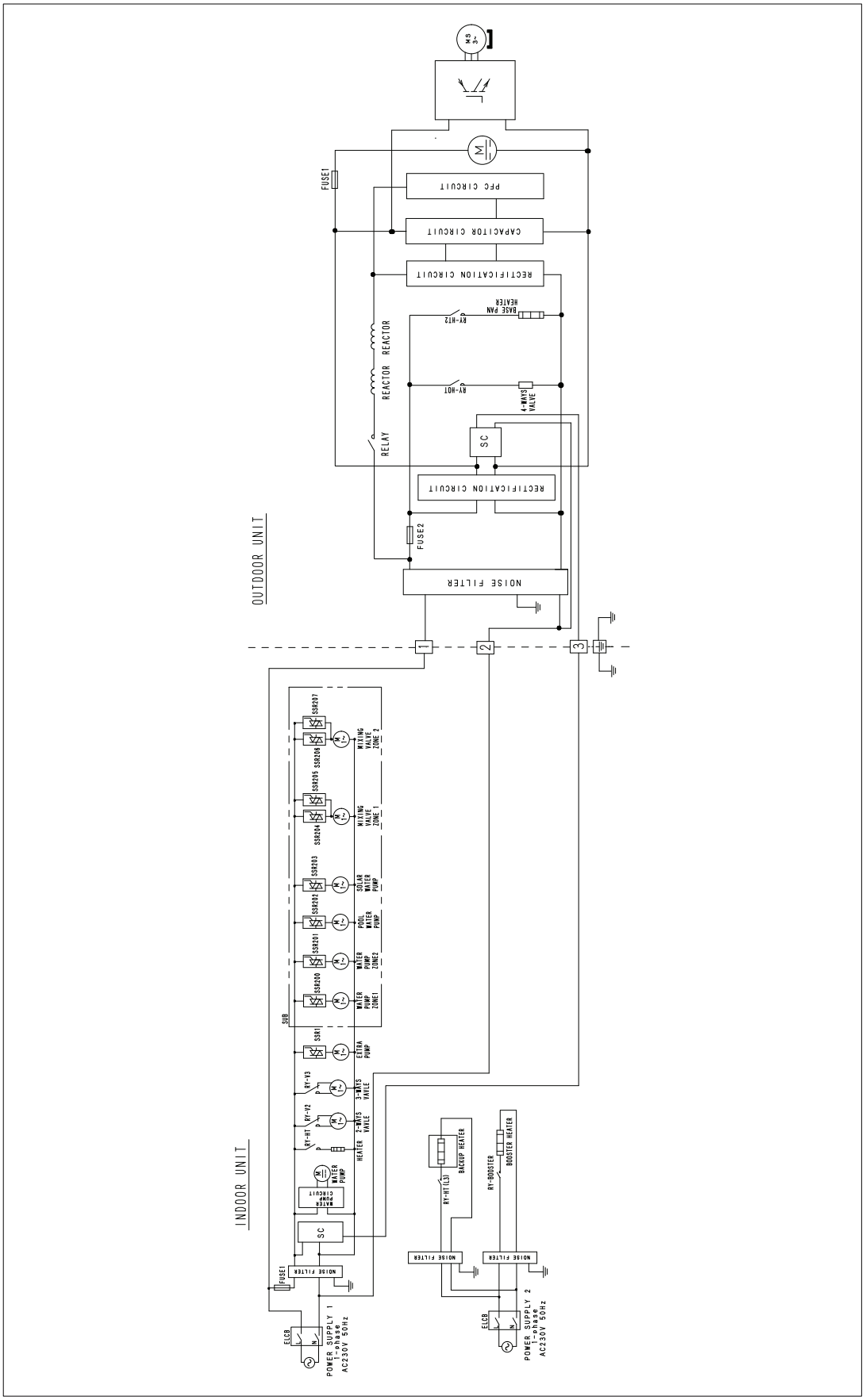


8. Block Diagram

8.1 WH-MDC05J3E5

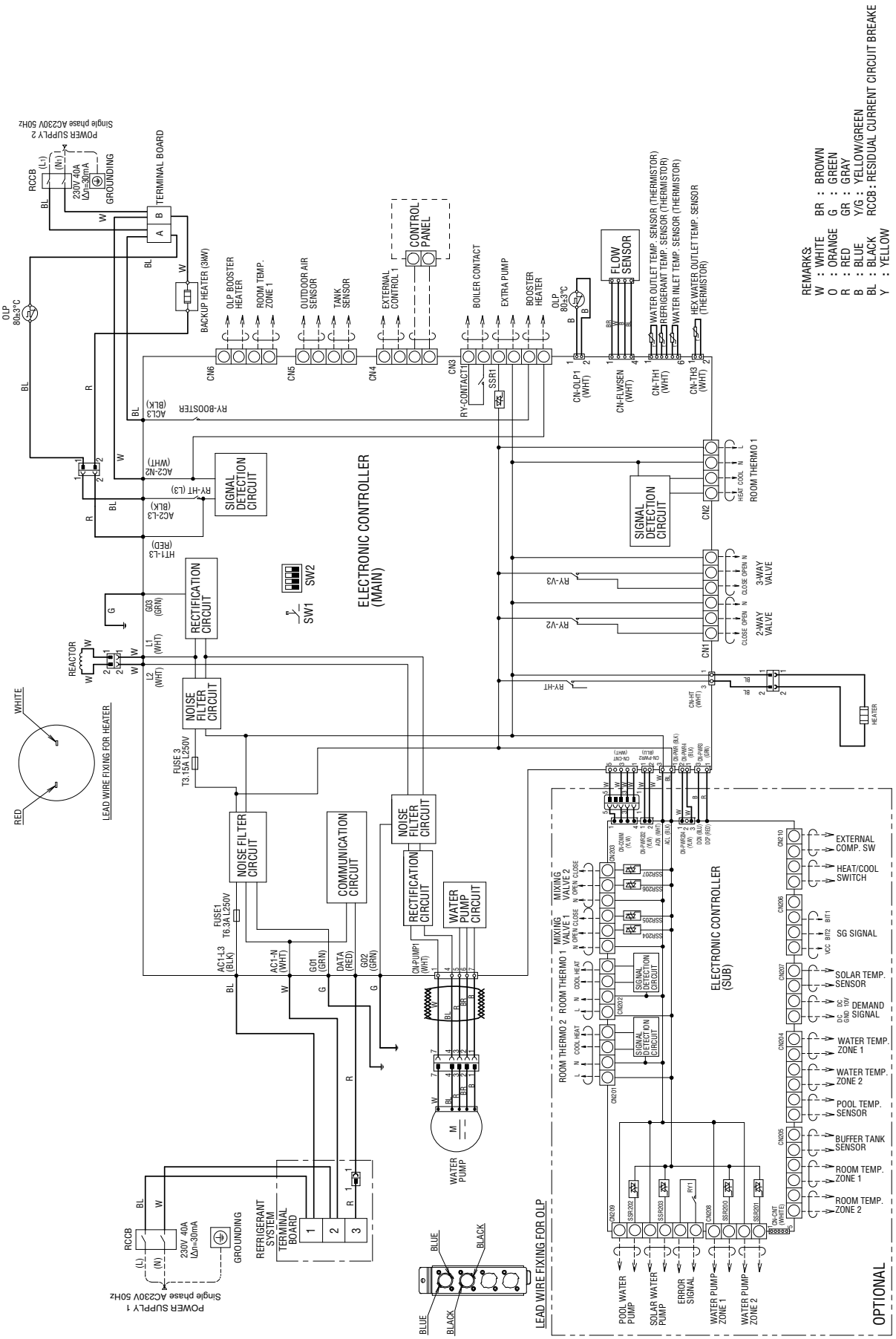


8.2 WH-MDC07J3E5 WH-MDC09J3E5



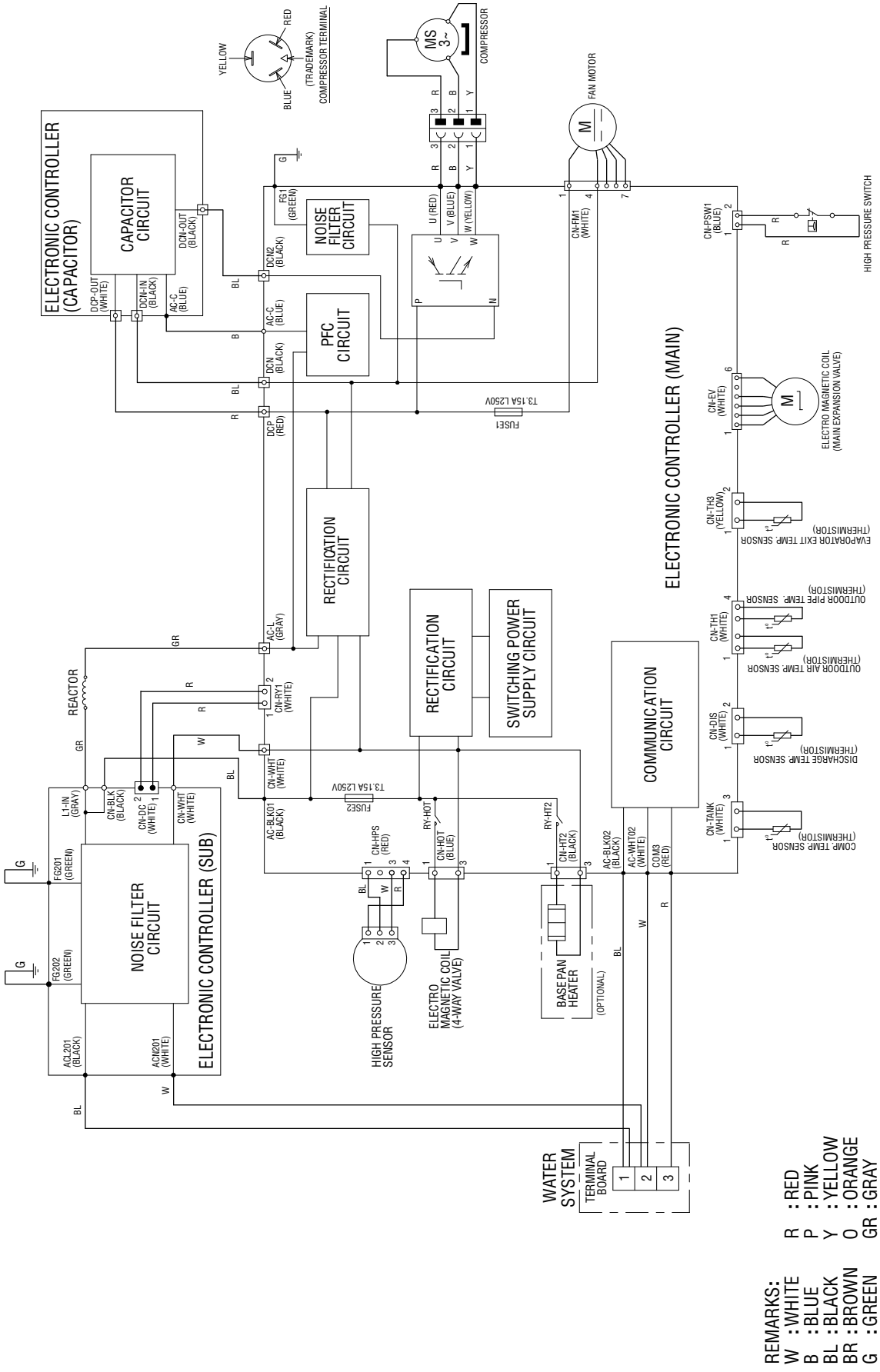
9. Wiring Connection Diagram

9.1 Water System

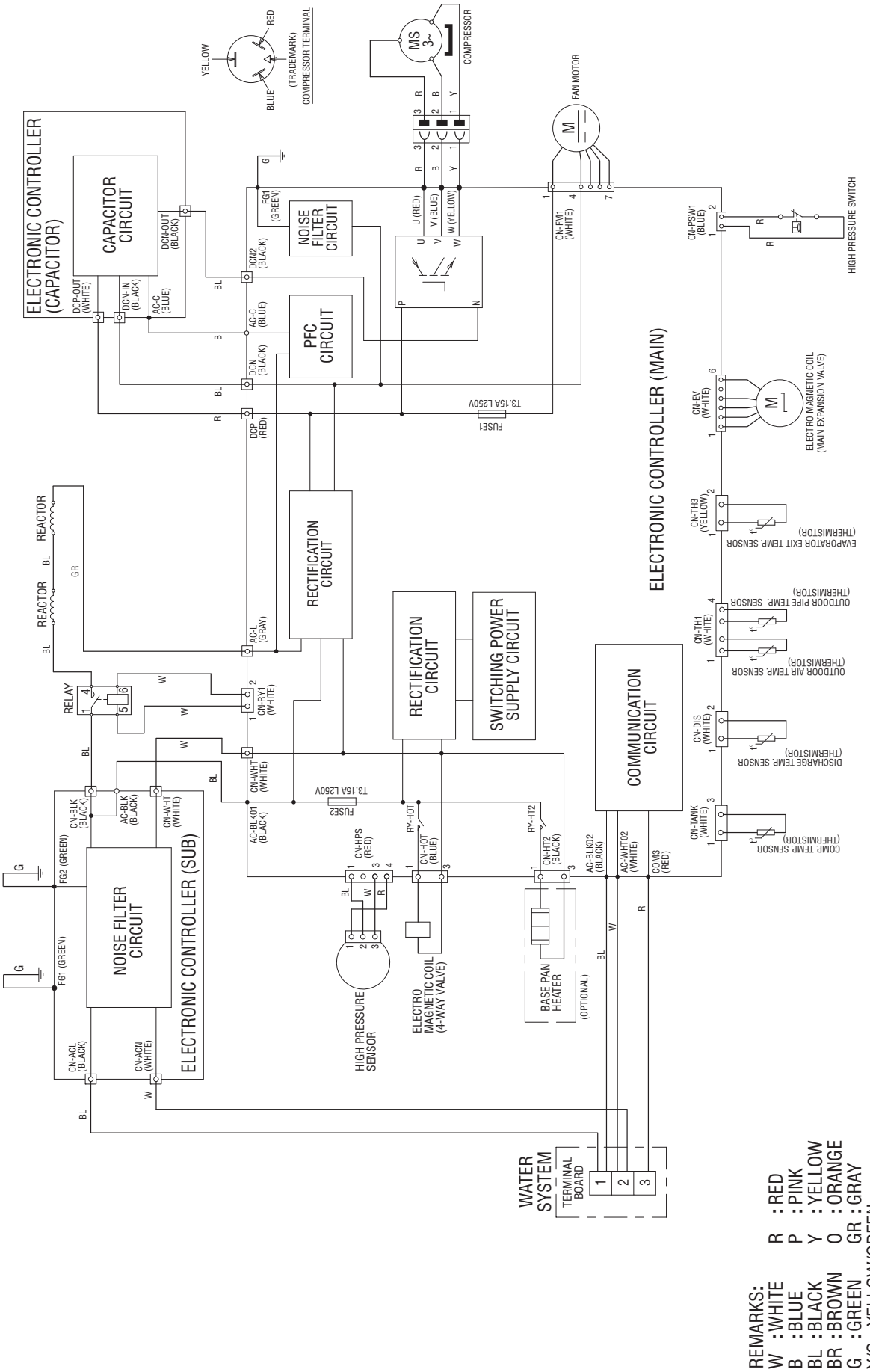


9.2 Refrigerant System

9.2.1 WH-MDC05J3E5

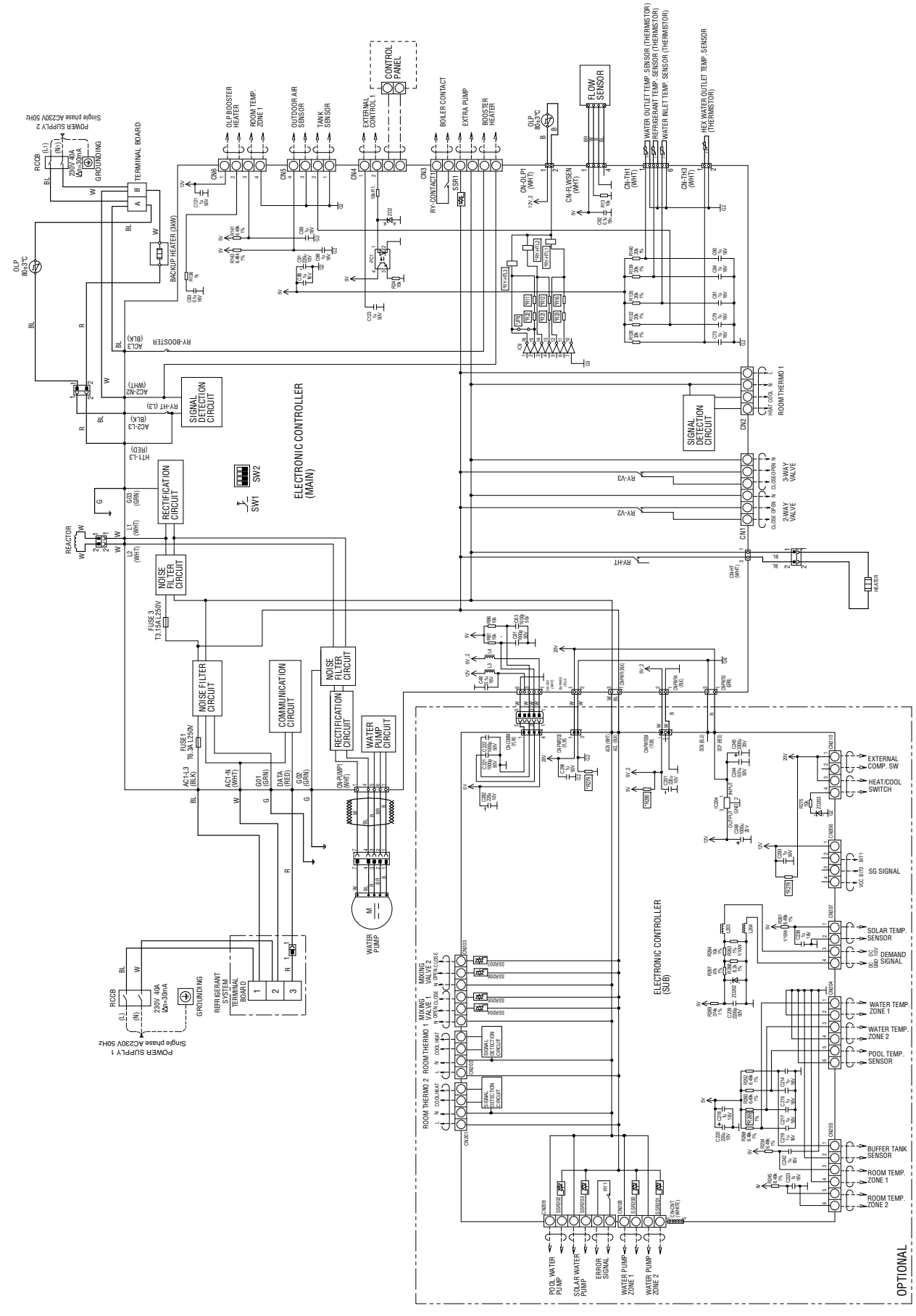


9.2.2 WH-MDC07J3E5 WH-MDC09J3E5



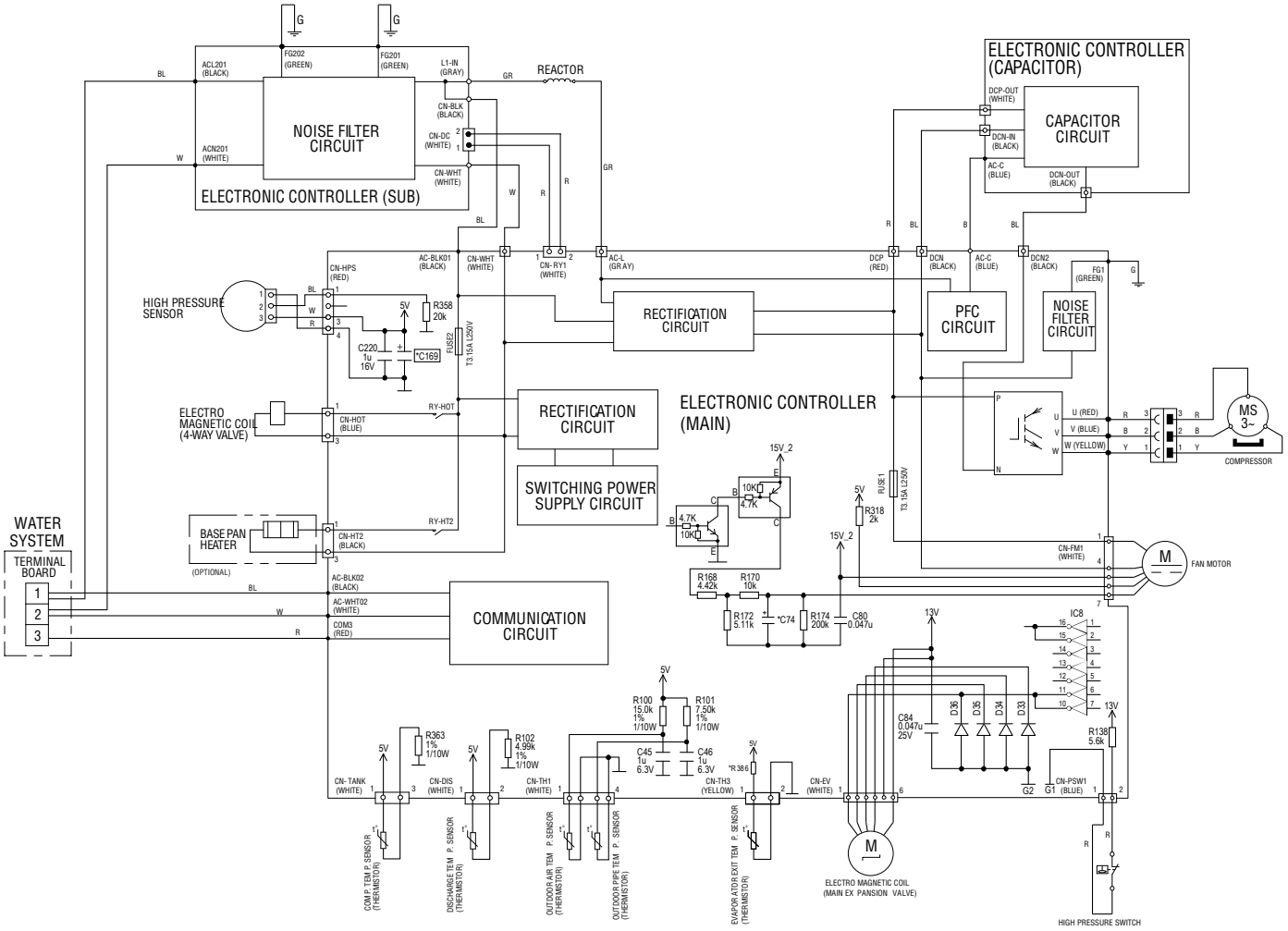
10. Electronic Circuit Diagram

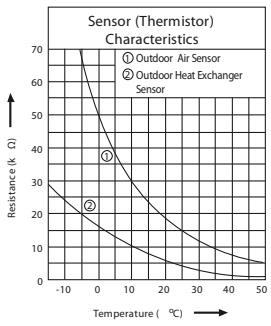
10.1 Water System

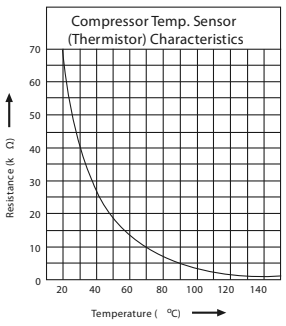


10.2 Refrigerant System

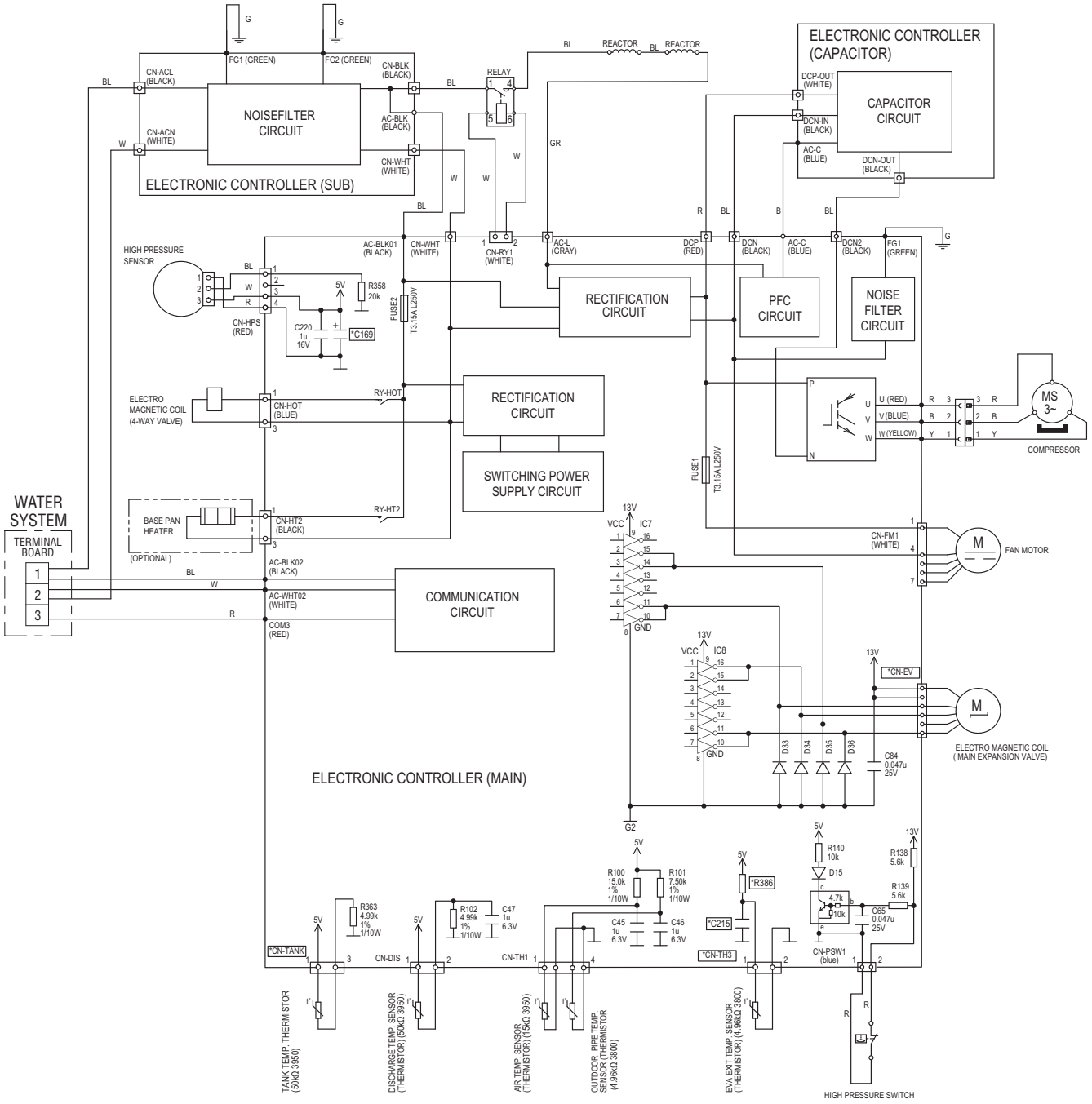
10.2.1 WH-MDC05J3E5

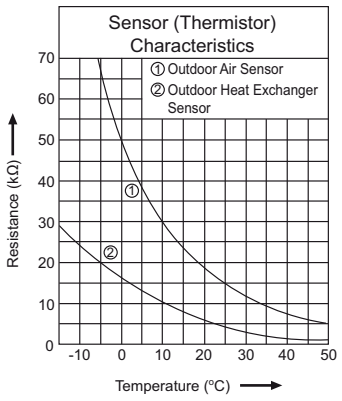


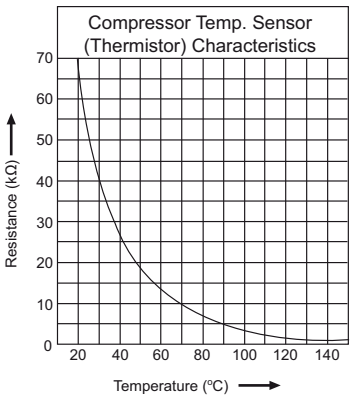




10.2.2 WH-MDC07J3E5 WH-MDC09J3E5



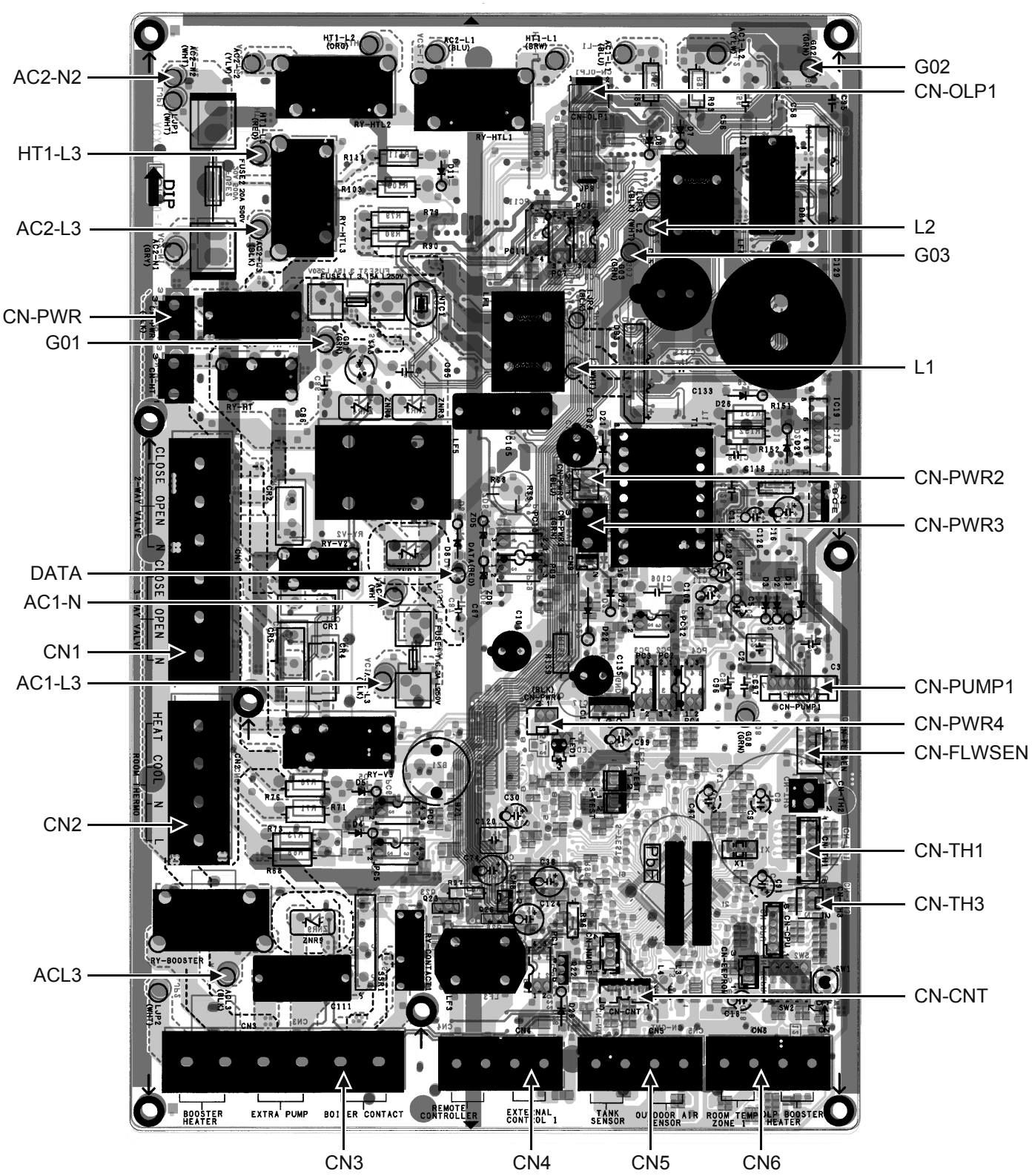




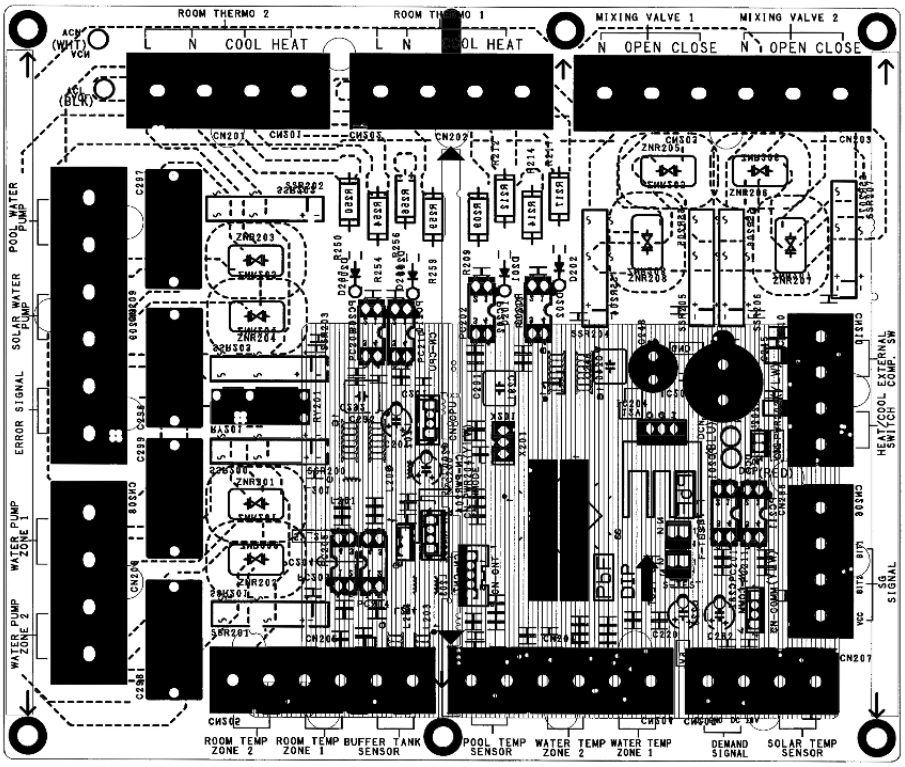
11. Printed Circuit Board

11.1 Water System

11.1.1 Main Printed Circuit Board

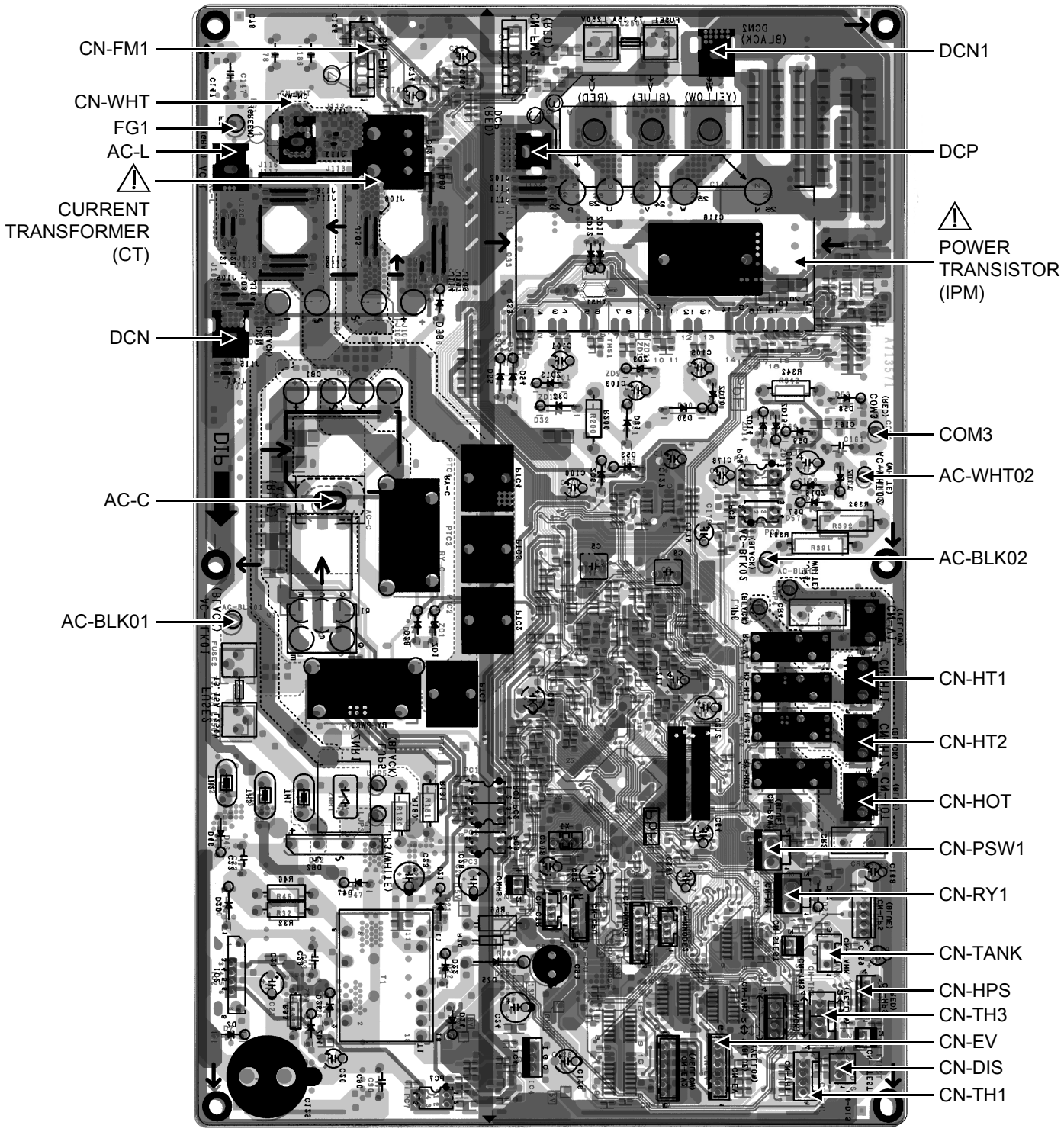


11.1.2 Optional Printed Circuit Board



11.2 Refrigerant System

11.2.1 Main Printed Circuit Board

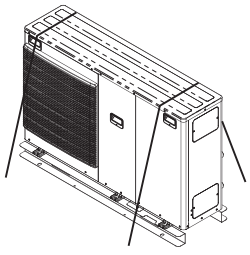


12. Installation Instruction

12.1 Mono Bloc Unit

12.1.1 Select the Best Location

- Install the Mono bloc unit in outdoor locations only.
- Avoid installations in areas where the ambient temperature may drop below -20°C.
- The Mono bloc unit must be installed on a flat, solid surface.
- A place removed from any heat source or steam which may effect the operation of the Mono bloc unit.
- A place where air circulation is good.
- A place where drainage can be easily done.
- A place where Mono bloc unit's operation noise will not cause discomfort to the user.
- A place which is accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated above from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- A place where the Mono bloc unit's piping and wiring lengths come within reasonable ranges.
- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- Avoid installing the Mono bloc unit at a location where suction side may be exposed directly to wind.
- If Mono bloc unit installed near sea, region with high content of sulphur or oily location (e.g. machinery oil, etc.), it lifespan maybe shorten.
- When installing the product in a place where it will be affected by typhoon or strong wind such as wind blowing between buildings, including the rooftop of a building and a place where there is no building in surroundings, fix the product with an overturn prevention wire, etc. (Overturn prevention fitting model number: K-KYZP15C)



12.1.2 Mono Bloc Unit Installation

Mono bloc unit will become heavy when filled with water. Please install the unit on a strong concrete floor and consider the weight of the unit and water.

- Fix Mono bloc unit on the concrete floor with M12 anchor bolt at 4 locations.
- Pull-out strength of these anchor bolts must be above 15000N.

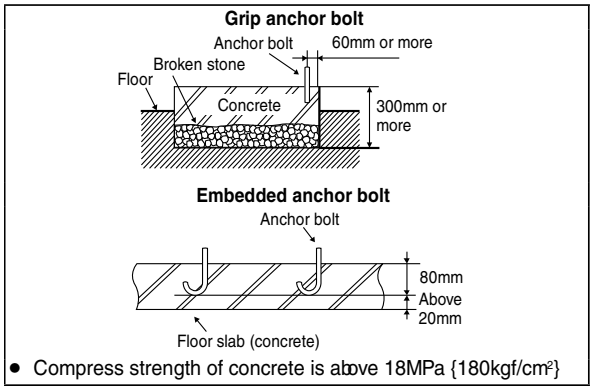
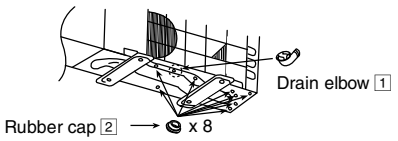


Illustration of grip type and embedded type anchor bolt

12.1.2.1 Disposal of Mono Bloc Unit Drain Water

- When a Drain elbow is used, please ensure to follow below:
  - The unit should be placed on a stand which is taller than 50mm.
  - Cover the 8 holes (ø20mm) with Rubber cap (refer to illustration below)
  - Use a tray (field supply) when necessary to dispose the Mono bloc unit drain water.

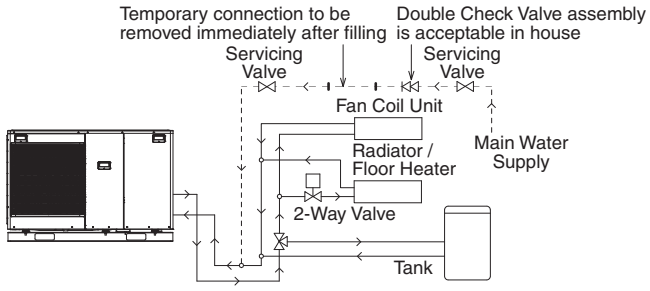


- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow and Rubber cap, for the drain water freezes and the fan will not rotate.

12.1.3 Piping Installation

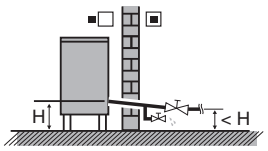
⚠ WARNING
This section is for authorized and licensed electrician / water system installer only. Work behind the front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

12.1.3.1 Typical Piping Installation



Please engage a licensed water circuit installer to install this water circuit.

- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not apply excessive force to piping that may damage the pipes.
- Use Rp 1¼" nut for both water inlet and water outlet connection and clean all piping with tap water before connecting to the Mono bloc unit.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall. If an existing tank is to be connected to this Mono bloc unit, ensure the pipes are clean before water pipe installation is carried out.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Tighten the nuts with torque wrench: 117.6 N•m.
- If non-brass metallic piping is used for installation, make sure to insulate the piping to prevent galvanic corrosion.
- Do not use pipes that are crushed or deformed. If these inferior pipes are used, it may cause unit malfunction.
- Make sure to insulate the water circuit piping (insulator thickness : 20mm or more) to prevent condensation during cooling operation and reduction of heating capacity, as well as avoid freezing of the outdoor water circuit piping during winter season.
- After installation, check the water leakage condition in connection area during test run.
- In case of a power supply failure or pump operating failure, drain the system (as suggested in the figure below).



When water is idle inside the system, freezing up is very likely to happen which could damage the system.

Drainage piping installation

- Use a drain hose with inner diameter of 15 mm.
- The hose must be installed in a continuously downward direction and left open to the frost-free atmosphere.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulfuric gas, etc.
- If necessary, use a hose clamp to tighten the hose at drain hose connector to prevent it from leaking.

⚠ CAUTION
Do not over tighten, over tightening cause water leakage.

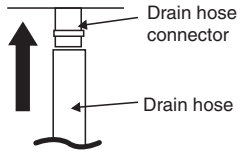


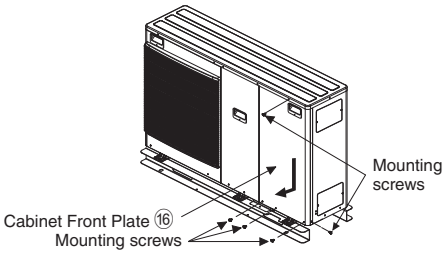
Illustration of how to fix drain hose to Mono bloc unit

12.1.4 Connect the Cable to Mono Bloc Unit

⚠ WARNING
This section is for authorised and licensed electrician only. Work behind the cabinet front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

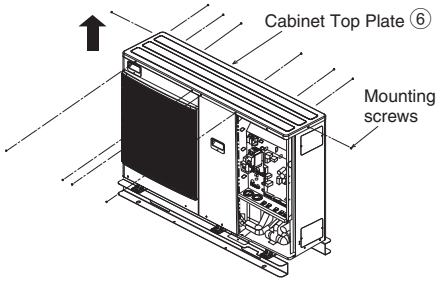
12.1.4.1 Remove The Cabinet Front Plate

1. Remove the 5 mounting screws as shown in the illustration.
2. Slide the Cabinet Front Plate downward to release the pawls.
   Then, pull it toward front to remove it.



12.1.4.2 Remove The Cabinet Top Plate

1. Remove the 12 mounting screws as shown in the illustration.
2. Lift the Cabinet Top Plate upward to remove it.



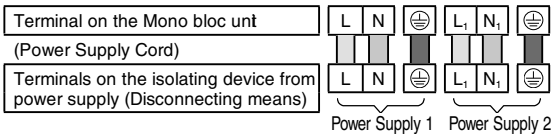
12.1.4.3 Fixing of Power Supply Cord

(REFER TO WIRING DIAGRAM AT UNIT FOR DETAIL)

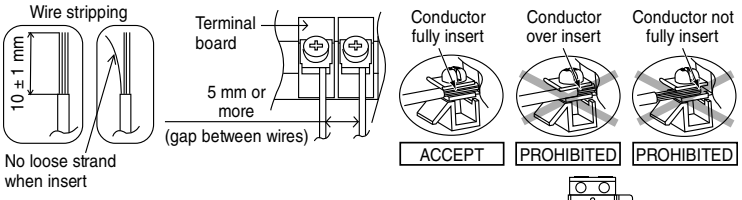
1. An isolating device must be connected to the power supply cable.
   - Isolating device (Disconnecting means) should have minimum 3.0 mm contact gap.
   - Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means). See below table for cable size requirement.

Model	Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
WH-MDC05J3E5	1	3 x min 1.5 mm^2	15/16A	30mA, 2P, type A
	2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC
WH-MDC07J3E5 and WH-MDC09J3E5	1	3 x min 2.5 mm^2	25A	30mA, 2P, type A
	2	3 x min 1.5 mm^2	15/16A	30mA, 2P, type AC

2. To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through the designated holes before being connected to the terminal block.
3. Secure the cable onto the control board with the holder (clamper).

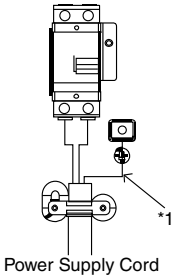


12.1.4.4 Wire Stripping and Connecting Requirement



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25}

*1 - Earth wire must be longer than other cables for safety reasons.



12.1.4.5 Connecting Requirement

For WH-MDC05J3E5

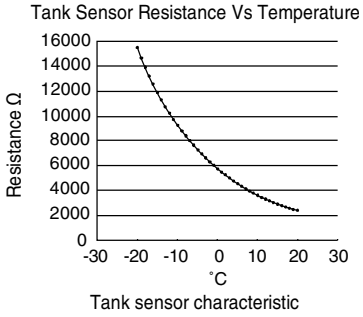
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-3 and can be connected to current supply network.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-11 and shall be connected to suitable supply network, with the following maximum permissible system impedance Z_{max} = 0.450 ohm (Ω) at the interface. Please liaise with supply authority to ensure that the Power Supply 2 is connected only to a supply of that impedance or less.

For WH-MDC07J3E5 and WH-MDC09J3E5

- This equipment's Power Supply 1 complies with IEC/EN 61000-3-12.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-11 and shall be connected to a suitable supply network, having services current capacity ≥ 100A per phase. Please liaise with supply authority that the service current capacity at the interface point is sufficient for the installation of the equipment.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-11 and shall be connected to suitable supply network, with the following maximum permissible system impedance Z_{max} = 0.450 ohm (Ω) at the interface. Please liaise with supply authority to ensure that the Power Supply 2 is connected only to a supply of that impedance or less.

12.1.4.6 Connecting with external device (optional)

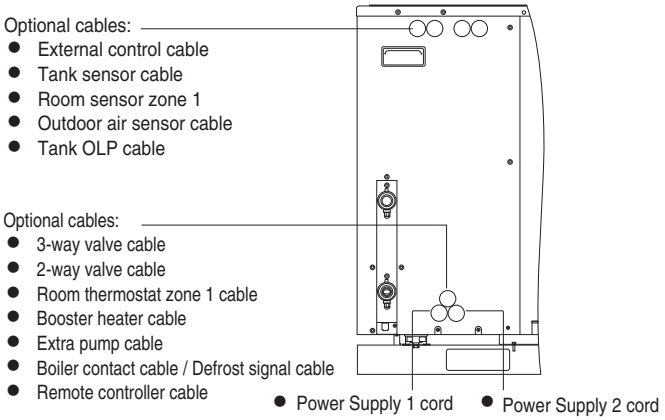
- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB
  1 Two-way valve shall be spring and electronic type, refer to "Field Supply Accessories" table for details. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
  * note: - Two-way Valve shall be CE marking compliance component.
  - Maximum load for the valve is 9.8VA.
  2 Three-way valve shall be spring and electronic type. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
  * note: - Shall be CE marking compliance component.
  - It shall be directed to heating mode when it is OFF.
  - Maximum load for the valve is 9.8VA.
  3 Room thermostat cable must be (4 or 3 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
  4 Maximum output power of booster heater shall be ≤ 3 kW. Booster heater cable must be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  5 Extra pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  6 Boiler contact cable shall be (2 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  7 External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  * note: - Switch used shall be CE compliance component.
  - Maximum operating current shall be less than $3A_{rms}$.
  8 Tank sensor shall be resistance type, please refer to graph below for the characteristic and details of sensor. Its cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of min 30V) of PVC-sheathed or rubber-sheathed cable.



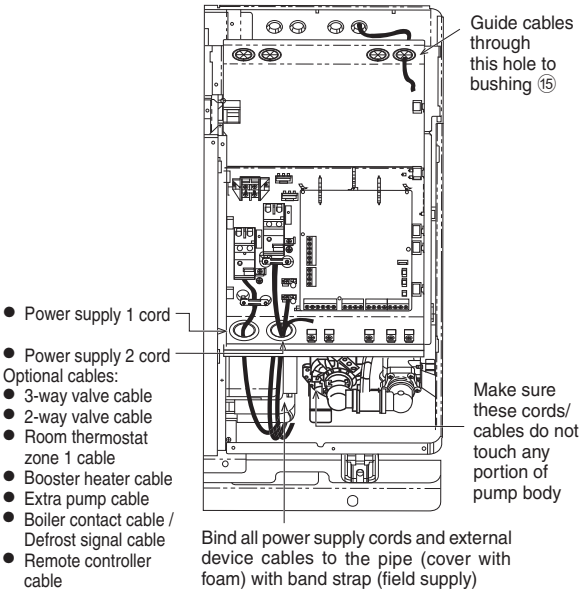
Tank sensor characteristic

  9 Room sensor zone 1 cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.

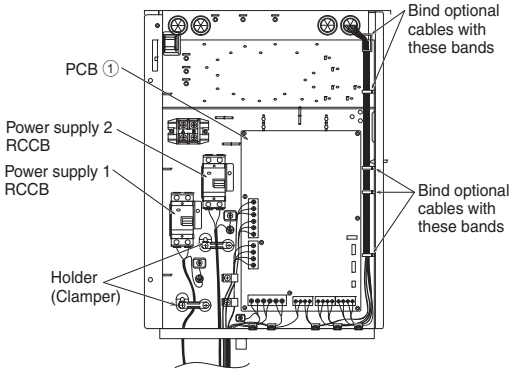
10 Outdoor air sensor cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.
11 Tank OLP cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide optional cables and power supply cords to Bushing ⑮



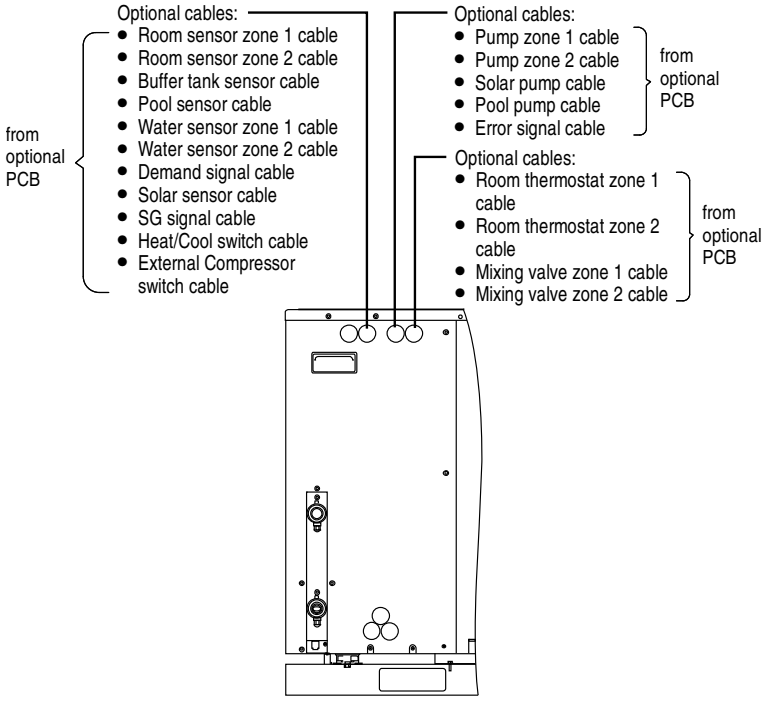
How to guide optional cables and power supply cords to control board



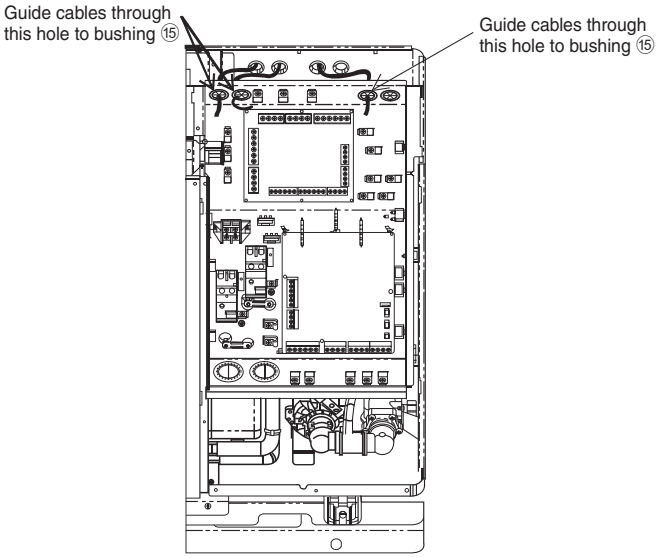
How to guide optional cables and power supply cords (view without internal wiring)

- For connection to optional PCB
  1 By connecting optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in optional PCB. Temperature of each zone can be controlled independently by remote controller.
  2 Pump zone 1 and zone 2 cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  3 Solar pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  4 Pool pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.

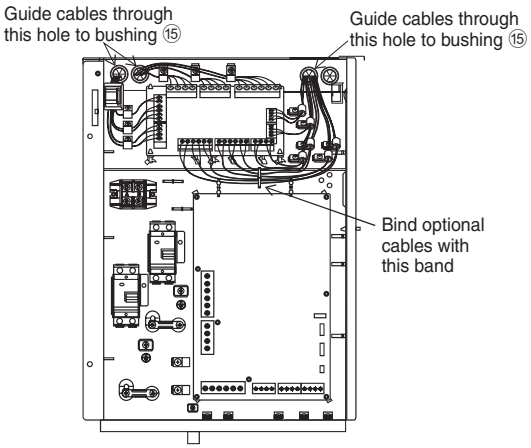
5 Room thermostat zone 1 and zone 2 cable shall be (4 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
6 Mixing valve zone 1 and zone 2 cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
7 Room sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
8 Buffer tank sensor, pool water sensor and solar sensor cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
9 Water sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
10 Demand signal cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
11 SG signal cable shall be (3 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
12 Heat/Cool switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
13 External compressor switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide optional cables to bushing ⑮



How to guide optional cables to control board



How to guide optional cables (view without internal wiring)

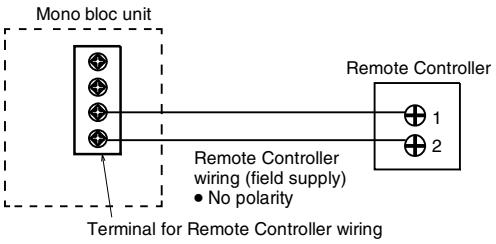
Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.1.5 Installation of Remote Controller

12.1.5.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1 By the window, etc. exposed to direct sunlight or direct air.
  2 In the shadow or backside of objects deviated from the room airflow.
  3 Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4 Location near heat source.
  5 Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

12.1.5.2 Remote Controller Wiring

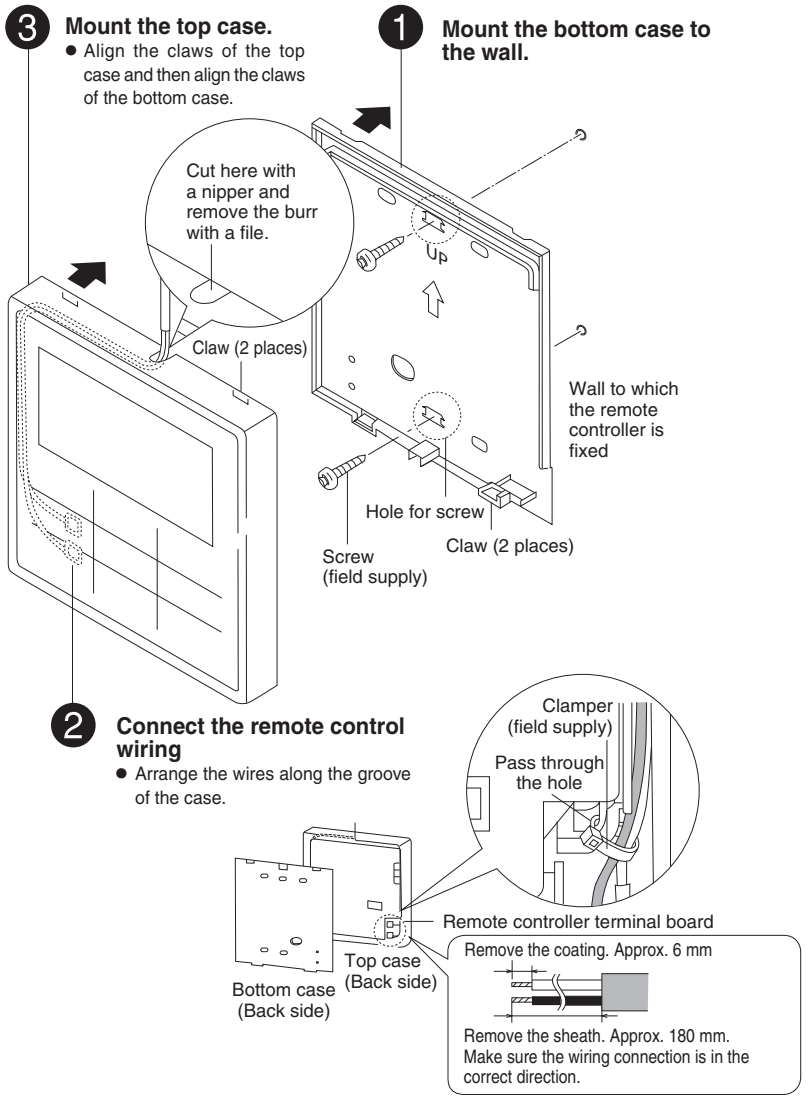


- Remote Controller cable shall be (2 x min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.

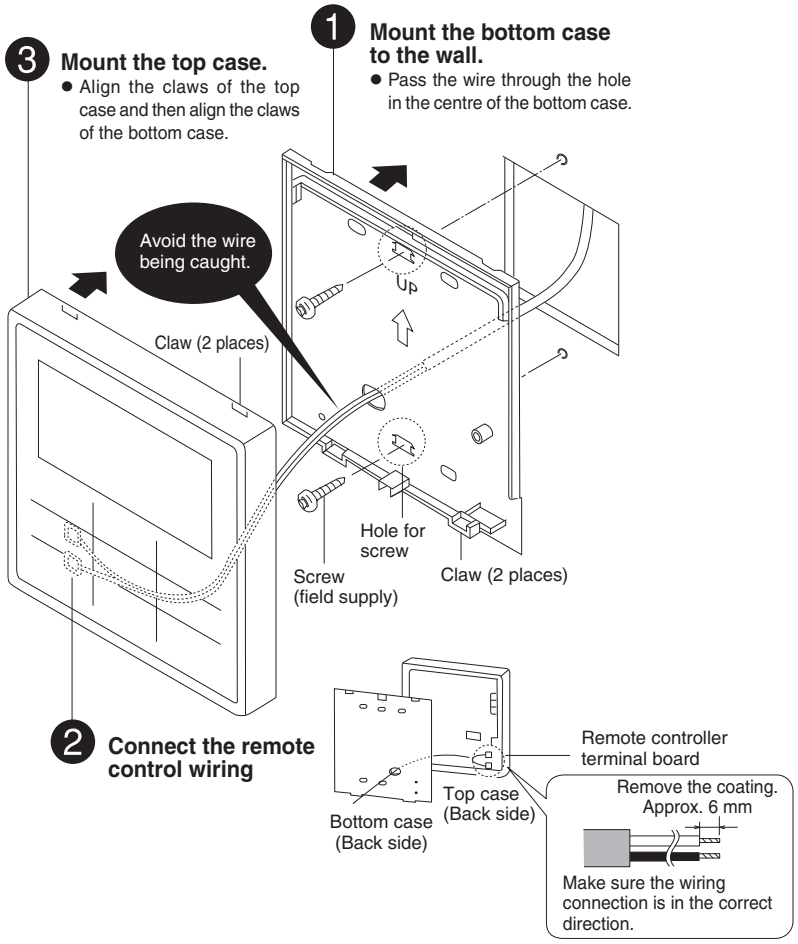
12.1.5.3 Mounting The Remote Controller

For exposed type

Preparation: Make 2 holes for screws using a driver.



For embedded type
Preparation: Make 2 holes for screws using a driver.

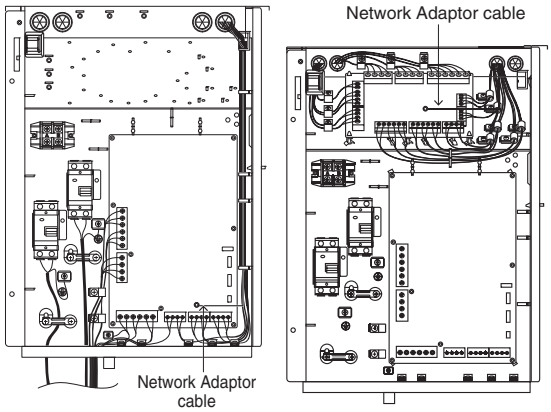


⚠ WARNING
This section is for authorized and licensed electrician only. Work behind the Cabinet Front Plate ⑯ secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

12.1.6 Installation of Network Adaptor and Base Pan Heater

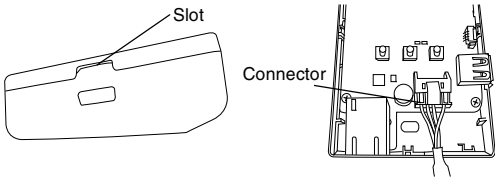
12.1.6.1 Network Adaptor and Installation (Optional)

1. **Open the Cabinet Front Plate and Cabinet Top Plate, then connect the Network Adaptor Cable to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Mono bloc unit so that there is no pinching.
   - If an optional PCB has been install in the Mono bloc unit, connect the CN-CNT connector to Optional PCB.

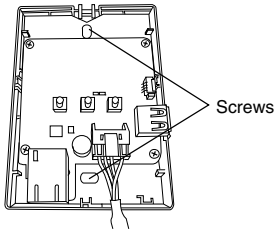


Without Optional PCB With Optional PCB

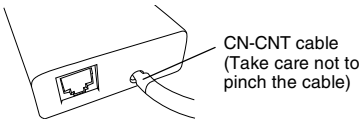
2. **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**



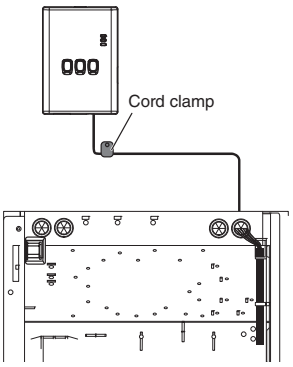
3. **On the wall near the Mono bloc unit, attach the adaptor by screwing screws through the holes in the back cover.**



4. **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**



5. **Use the included cord clamp to fix the CN-CNT cable to the wall.**
   Pull the cable around as shown in the diagram so that external forces cannot act on the connector in the adaptor.



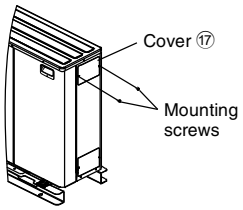
12.1.6.2 Base Pan Heater (Optional)

- It is strongly recommended to install a Base Pan Heater (optional) if the Mono bloc unit is installed in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.

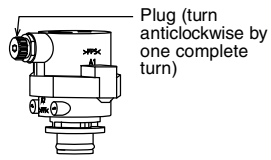
12.1.7 Charging the Water

- Make sure all the piping installations are properly done before carry out below steps.

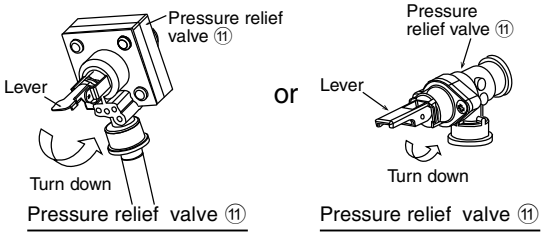
1 Take out the Cover by removing the 2 mounting screws to access to the Pressure Relief Valve and Air Purge Valve.



2 Turn the plug on the Air Purge Valve outlet anticlockwise by one complete turn from fully closed position.



3 Set the Pressure Relief Valve level “DOWN”.



4 Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Mono bloc unit via water inlet. Stop filling water if the free water flow through Pressure Relief Valve drain hose.
5 Turn ON the power supply and make sure Water Pump is running.
6 Check and make sure no water leaking at the tube connecting points.
7 Reinstall the Cover by tightening the 2 mounting screws.

12.1.8 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings. Before obtaining access to terminals, all supply circuits must be disconnected.

*12.1.8.1 Check Water Pressure *(0.1 MPa = 1 bar)*

Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure Gauge). If necessary add tap water into the water circuit.

12.1.8.2 Check Pressure Relief Valve

- Check for correct operation of Pressure Relief Valve by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keeps drained out from the unit, switch off the system, and then contact your local authorized dealer.

12.1.8.3 Expansion Vessel Pre Pressure Checking

[Upper limit water volume of the system]
The Mono bloc unit has a build-in Expansion Vessel with 6 L air capacity and initial pressure of 1 bar.

- Without antifreeze agent condition

Total amount of water in the system should be below 150 L.
If the total amount of water is more than 150 L, please add expansion vessel (field supply).

- With antifreeze agent condition

In the case of using antifreeze agent, expansion rate ε is different depending on its maker.
Please refer to the antifreeze agent maker for the expansion rate ε before calculate the upper limit water volume of the system.
The expansion vessel capacity required for the system can be calculated from the formula below.

$$V = \frac{\varepsilon \times Vo}{1 - \frac{98 + P_1}{98 + P_2}}$$

V : Required gas volume <expansion vessel volume L>
Vo : System total water volume <L>
ε : Expansion rate 5→60˚C = (depends on antifreeze agent used)
P_1 : Expansion tank filling pressure = (100) kPa
P_2 : System maximum pressure = 300 kPa
- ( ) Please confirm at actual place
- The gas volume of the sealed type expansion vessel is presented by <V>.

❍ It’s advised to add 10% margin for required gas volume of calculation.

[Adjustment of the initial pressure of the expansion vessel when there is a difference in installation height]
If the height difference between the Mono bloc unit and the highest point of the system water circuit (H) is more than 7m, please adjust the initial pressure of the expansion vessel (Pg) according to the following formula.

Pg= (H*10+30) kPa

12.1.8.4 Check RCCB

Ensure the RCCB set to “ON” condition before check RCCB 2.
Turn on the power supply to the Mono bloc unit.
This testing could only be done when power is supplied to the Mono bloc unit.

⚠ WARNING
Be careful not to touch parts other than RCCB test button when the power is supplied to Mono bloc unit. Else, electrical shock may happen.

- Push the “TEST” button on the RCCB. The lever would turn down and indicate “0”, if it functions normal.
- Contact authorized dealer if the RCCB malfunction.
- Turn off the power supply to the Mono bloc unit.
- If RCCB functions normal, set the lever to “ON” again after testing finish.

This product contains fluorinated greenhouse gasses.
Refrigerant type : R32 (GWP=675)
Amount : 1.30 kg (0.878 ton CO_2 equivalent)

12.1.9 Test Run

1. Before test run, make sure below items have been checked:-
   - Pipework are properly done.
   - Electric cable connecting work are properly done.
   - Mono bloc unit is filled up with water and trapped air is released.
2. Antifreeze agent must be added into water circuit to prevent freezing of water when outdoor ambient temperatures is low. Recommended antifreeze: Propylene glycol: 40% (equivalent to -20°C)
3. Set ON to the Mono bloc unit and RCCB. Then, for remote control operation please refers to (Mono bloc) Air-to-Water Heatpump’s operation instruction.
4. For normal operation, pressure gauge reading should be in between 0.05 MPa and 0.3 MPa.
5. After test run, please clean the Water filter. Reinstall it after finish cleaning.

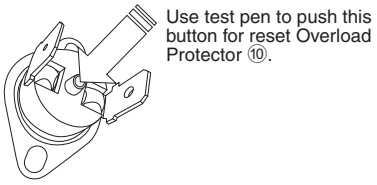
12.1.9.1 Check Water Flow of Water Circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]

12.1.9.2 Reset Overload Protector

Overload Protector serves the safety purpose to prevent the water over heating. When the Overload Protector trip at high water temperature, take below steps to reset it.

1. Take out the cover.
2. Use a test pen to push the centre button gently in order to reset the Overload Protector.
3. Fix the cover to the original fixing condition.



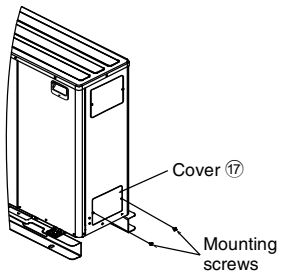
12.1.10 Maintenance

- In order to ensure optimal performance of the unit, seasonal inspections on the unit, functional check of RCCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer.

12.1.10.1 Maintenance for Water Filter Set

1. Remove the Cover by loosening the mounting screws to access to the Water Filter Set.
2. Turn OFF power supply.
3. Set the two valves for the Water Filter Set to “CLOSE”.
4. Take off the clip, then gently pull out the mesh. Beware of small amount water drain out from it.
5. Clean the mesh with warm water to remove all the stain. Use soft brush if necessary.
6. Reinstall the mesh to the Water Filter Set and set back the clip on it.
7. Set the two valves for the Water Filter Set to “OPEN”.
8. Turn ON power supply.
9. After cleaning, reinstall the Cover by tightening the mounting screws properly.

⚠ WARNING
Do not add or replace other than R32 type. It may cause product damage, burst, injury and etc. Use compatible R32 tools for refrigerant piping work and refrigerant charging during installation or servicing.



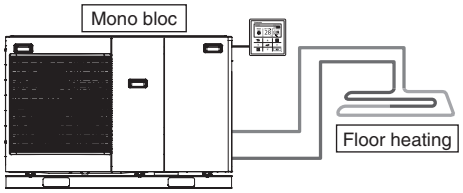
12.2 Variation of system

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.

12.2.1 Introduce application related to temperature setting.

12.2.1.1 Temperature setting variation for heating

1. Remote Controller



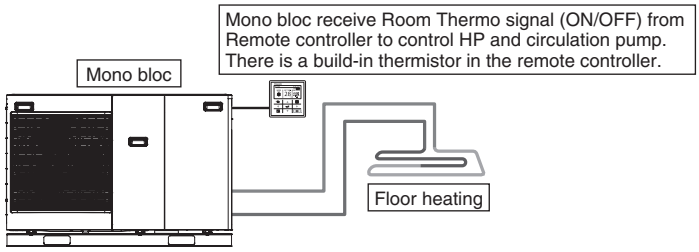
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Mono bloc.
This is the basic form of the most simple system.

2. Room Thermostat



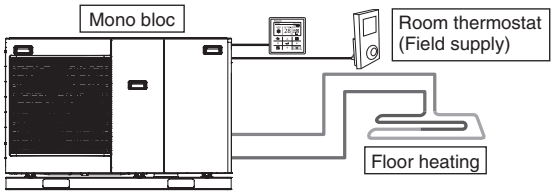
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

Connect floor heating or radiator directly to the Mono bloc.
Install the remote controller in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat



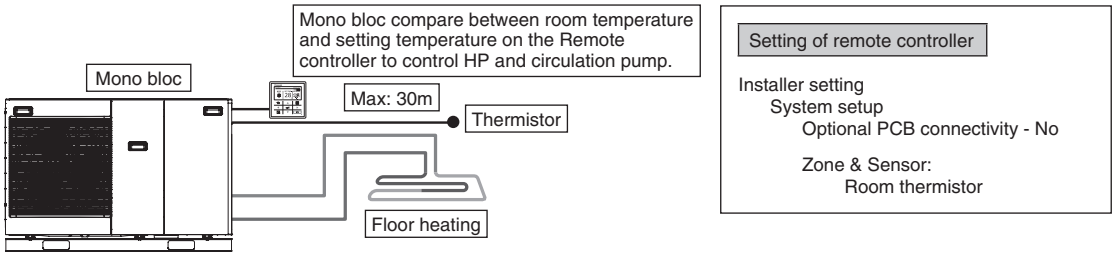
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Mono bloc.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



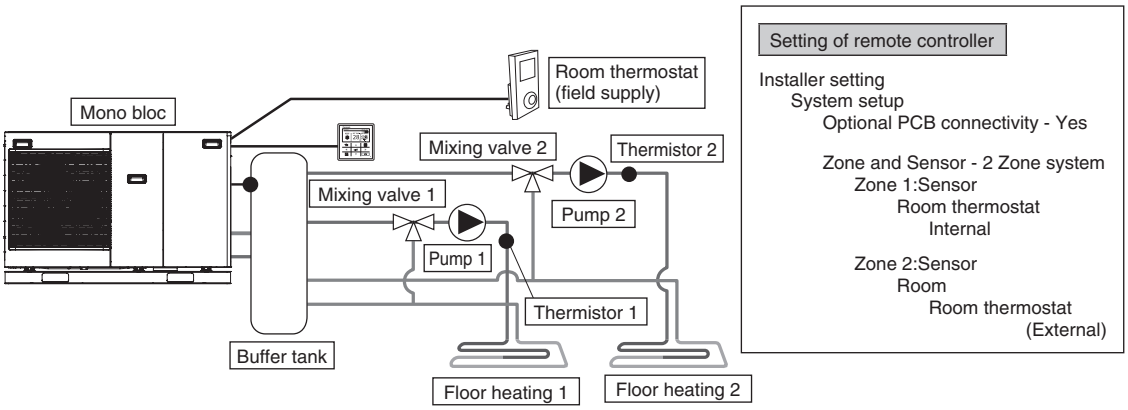
Connect floor heating or radiator directly to Mono bloc.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.

- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

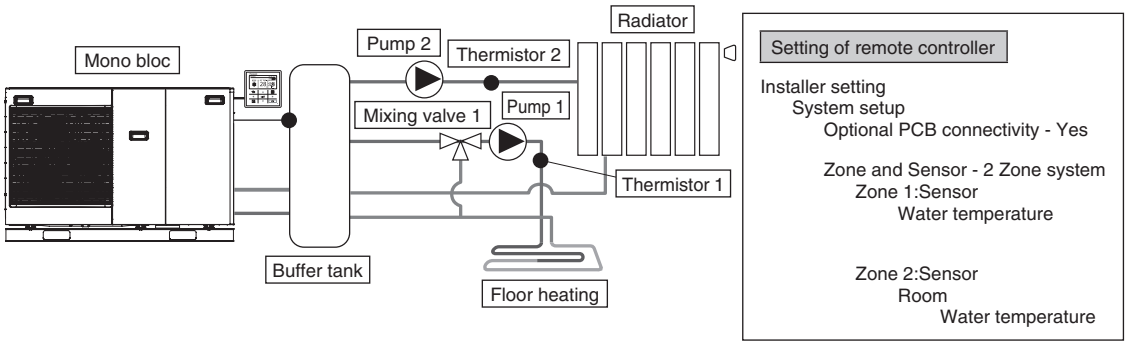
12.2.1.2 Examples of installations

Floor heating 1 + Floor heating 2



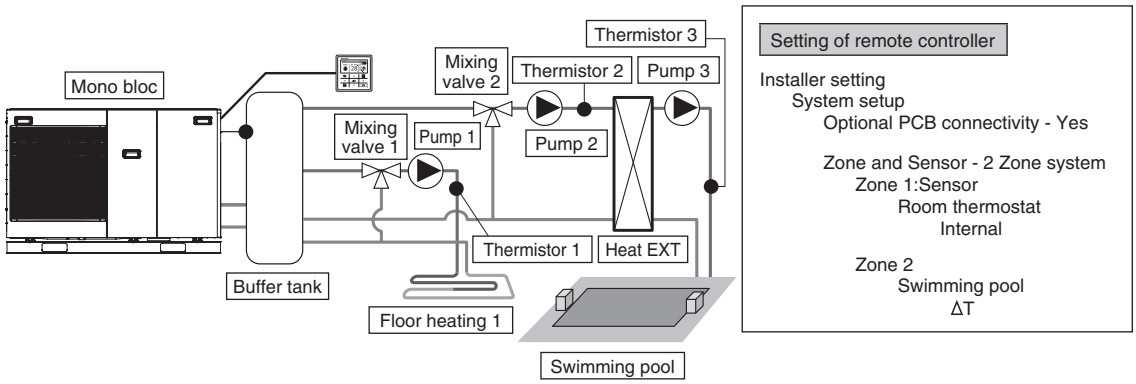
Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Install the remote controller in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires optional PCB (CZ-NS4P).

Floor heating + Radiator



Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the optional PCB (CZ-NS4P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.

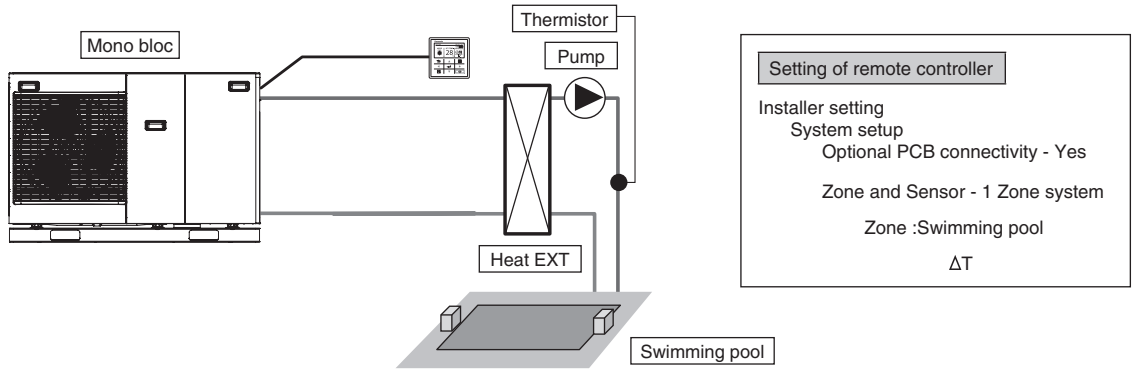
Floor heating + Swimming pool



Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Install the remote controller in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the optional PCB (CZ-NS4P).

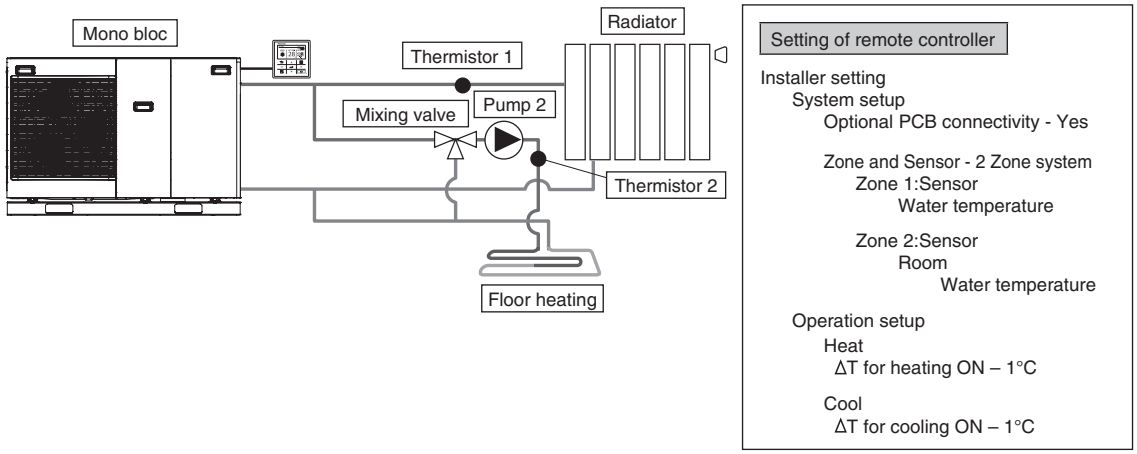
※ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.

Swimming pool only



This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Mono bloc without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Install the remote controller in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the optional PCB (CZ-NS4P).
In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



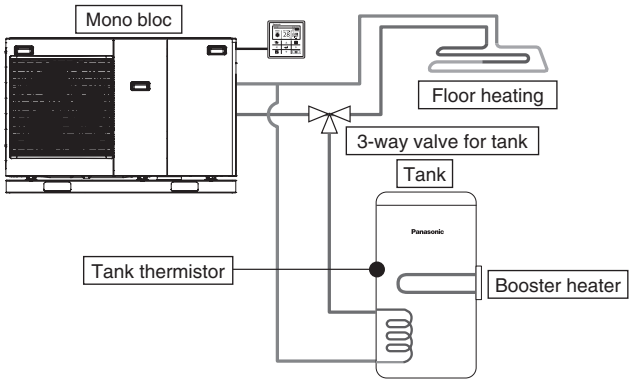
This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Mono bloc served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the optional PCB (CZ-NS4P).

(NOTE)

- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
  Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

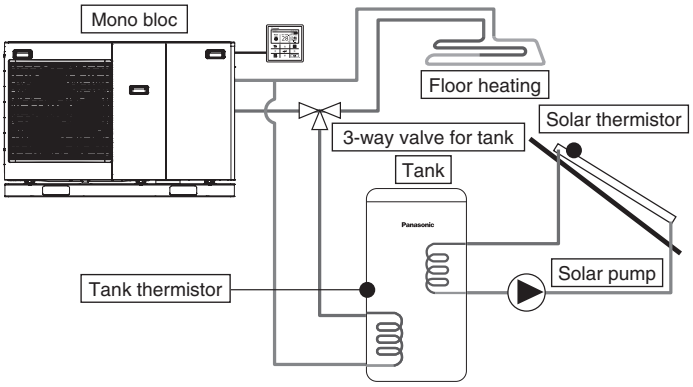
12.2.2 Introduce applications of system that uses optional equipment.

DHW (Domestic Hot Water) Tank connection



This is an application that connects the DHW tank to the Mono bloc through 3-way valve.
DHW tank's temperature is detected by tank thermistor (specified by Panasonic).

Tank + Solar connection

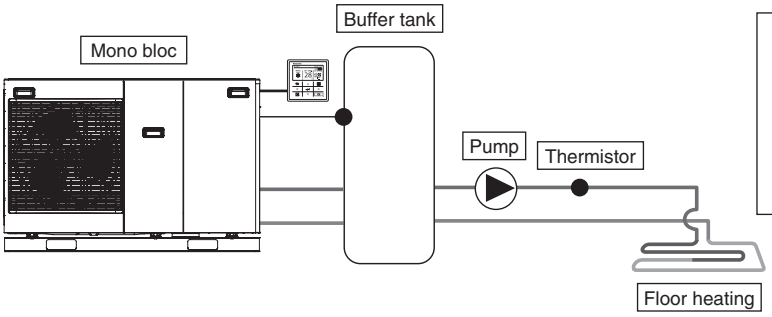


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Tank connection - Yes
Solar connection - Yes
DHW tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the DHW tank to the Mono bloc through 3-way valve before connect the solar water heater to heat up the tank. DHW tank's temperature is detected by tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
DHW tank shall use tank with built-in solar heat exchange coil independently.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
This system requires optional PCB (CZ-NS4P).

Buffer tank connection

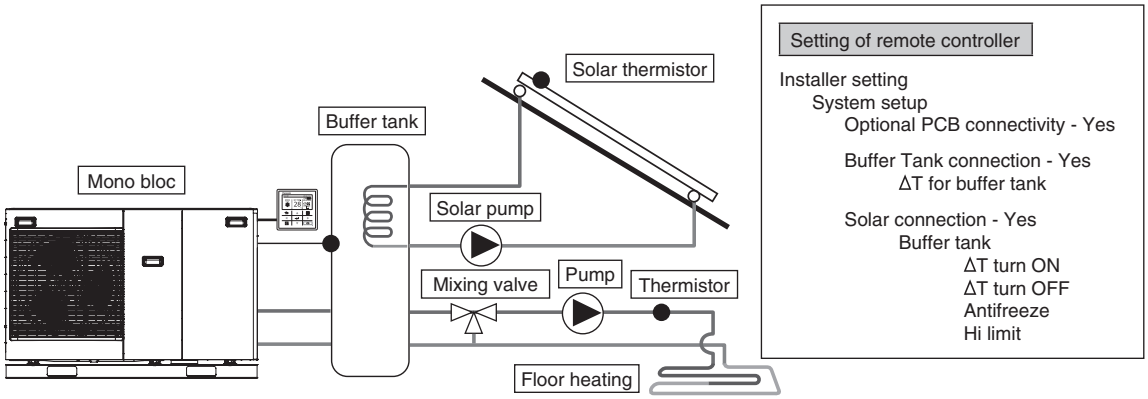


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank

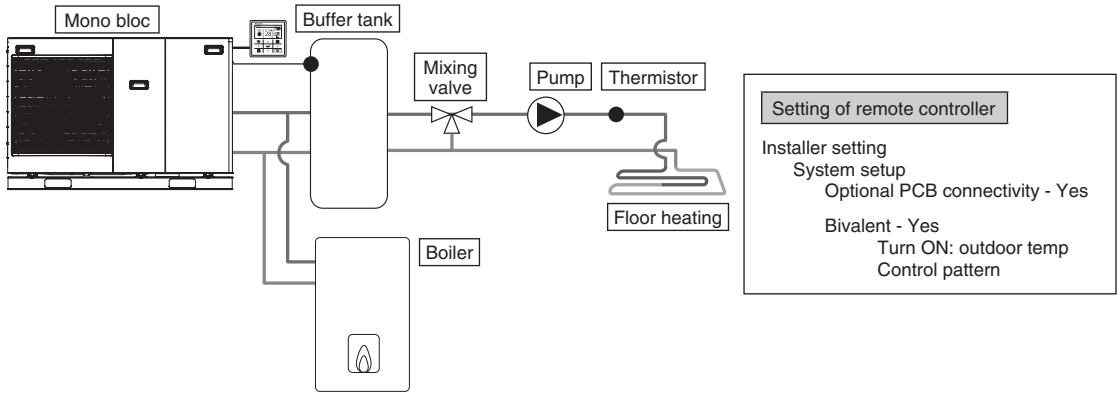
This is an application that connects the buffer tank to the Mono bloc.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
This system requires optional PCB (CZ-NS4P).

Buffer tank + Solar



This is an application that connects the buffer tank to the Mono bloc before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires optional PCB (CZ-NS4P).

Boiler connection



This is an application that connects the boiler to the Mono bloc, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
There are 3 modes selectable by remote controller for boiler connection.
Besides that, an application that connects to the DHW tank's circuit to heat up tank's hot water is also possible. (Operation setting of boiler shall be responsible by installer.)
This system requires optional PCB (CZ-NS4P).
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when selecting Advanced Parallel setting.)

⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the indoor unit does NOT exceed 55°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

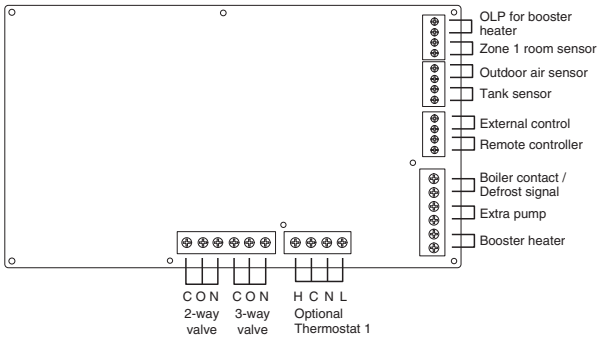
12.3 How to Fix External Device

12.3.1 Connecting Cables Length

When connecting cables between Mono bloc and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Three-way valve	50
Mixing valve	50
Room thermostat	50
Booster heater	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Tank sensor	30
Room sensor	30
Outdoor air sensor	30
Tank OLP	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.3.1.1 Connection of the main PCB



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal ❊It does not function when using the optional PCB
OLP for booster heater	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) It is connected to the safety device (OLP) of DHW tank.
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

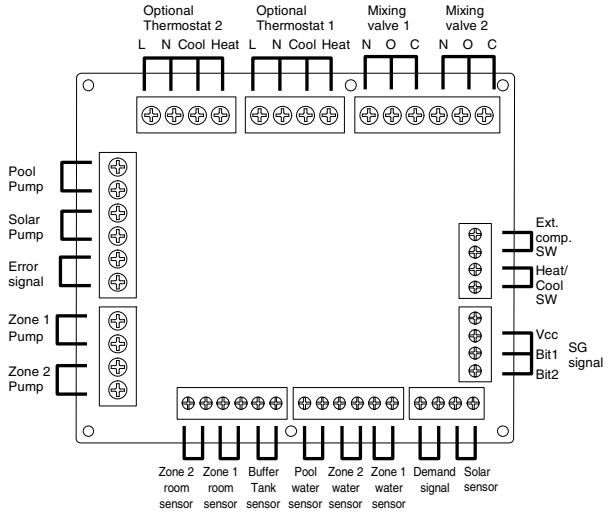
■ Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)
Extra pump	AC230V (Used when Mono bloc pump capacity is insufficient)
Booster heater	AC230V (Used when using booster heater in DHW tank)
Boiler contact/ Defrost signal	Dry contact (System setup necessary)

■ Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ❊It does not work when using the optional PCB
Outdoor air sensor	AW-A2W-TSOD (Total cable length shall be 30m or less)
Tank sensor	Please use Panasonic specified part

12.3.1.2 Connection of Optional PCB (CZ-NS4P)



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.ON, Short=Comp.OFF (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

■ Outputs

Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s
Pool pump	AC230V
Solar pump	AC230V
Zone pump	AC230V

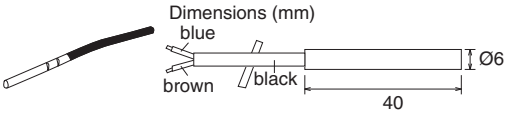
■ Thermistor inputs

Zone room sensor	PAW-A2W-TSRT
Buffer tank sensor	PAW-A2W-TSBU
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

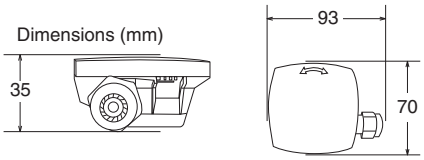
12.3.1.3 Recommended External Device Specification

- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.

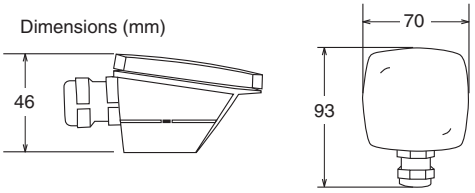
1 Buffer tank sensor: PAW-A2W-TSBU
Use for measurement of the buffer tank temperature.
Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



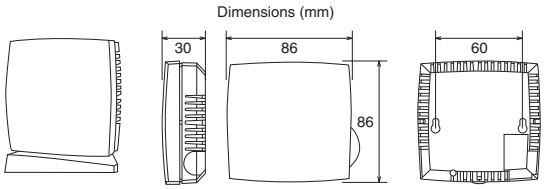
2 Zone water sensor: PAW-A2W-TSHC
Use to detect the water temperature of the control zone.
Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



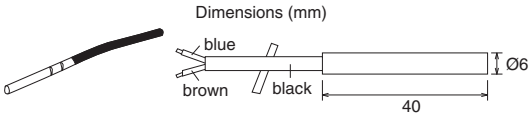
3 Outdoor sensor: PAW-A2W-TSOD
If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



4 Room sensor: PAW-A2W-TSRT
Install the room temperature sensor to the room which requires room temperature control.



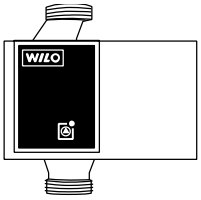
5 Solar sensor: PAW-A2W-TSSO
Use for measurement of the solar panel temperature.
Insert the sensor into the sensor pocket and paste it on the solar panel surface.



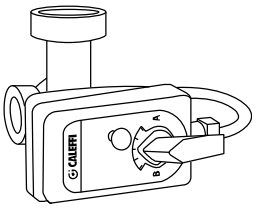
6 Please refer to the table below for sensor characteristic of the sensors mentioned above.

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo

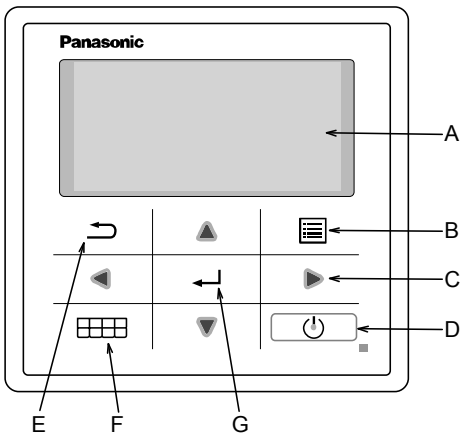


- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

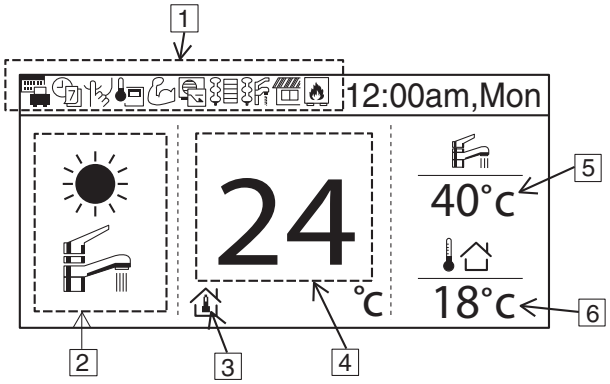


12.4 System Installation

12.4.1 Remote Controller Outline



	Name	Function
A:	Main screen	Display information
B:	Menu	Open/Close main menu
C:	Triangle (Move)	Select or change item
D:	Operate	Start/Stop operation
E:	Back	Back to previous item
F:	Quick Menu	Open/Close Quick menu
G:	OK	Confirm



	Name	Function
1:	Function icon	Display set function/status

Holiday mode, Weekly timer, Quiet mode, Remote controller room thermostat, Powerful mode, Demand control, Room heater, Tank heater, Solar, Boiler

2: Mode — Display set mode/current status of mode

Heating, Cooling, Auto, Hot water supply, Auto heating, Auto cooling, Heat pump operating

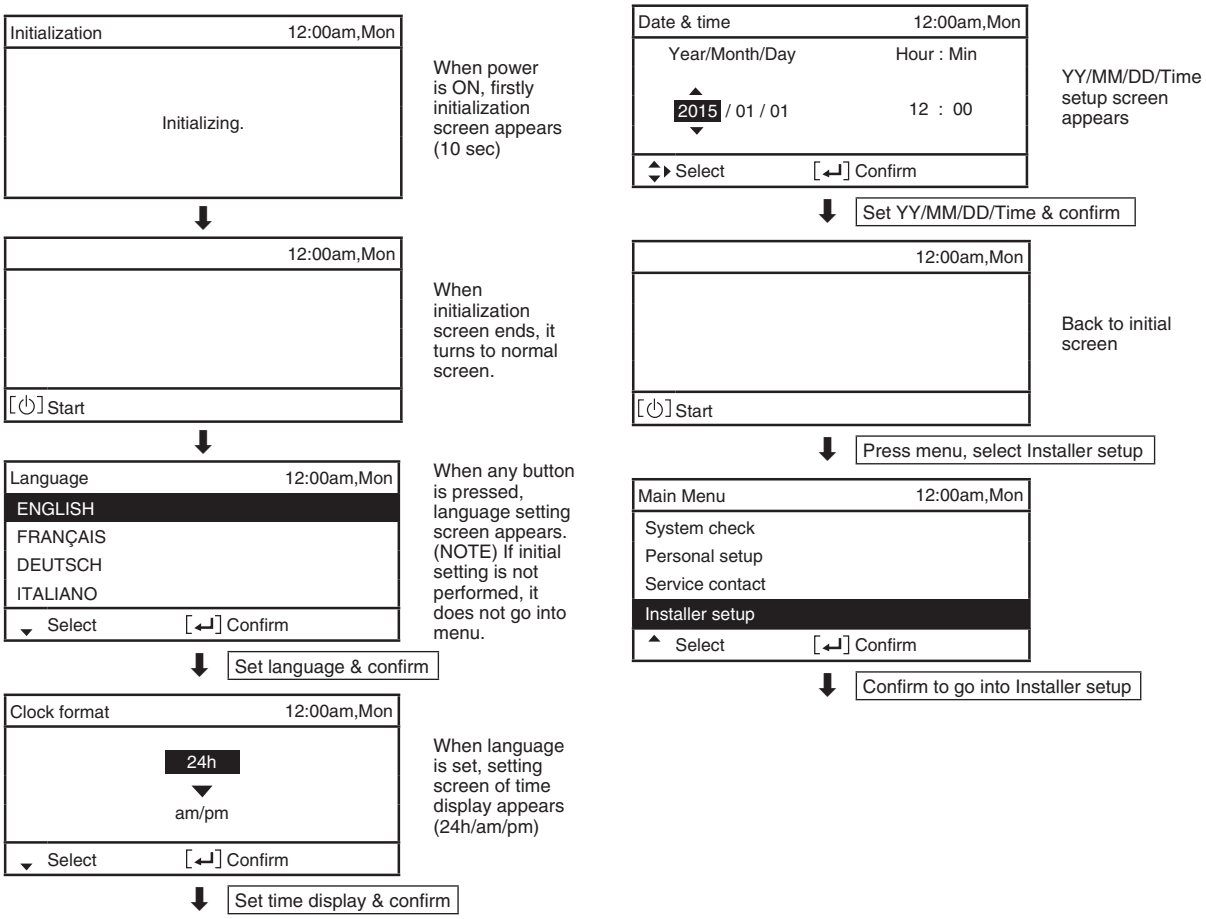
3: Temp setting — Set room temp, Compensation curve, Set direct water temp, Set pool temp

4: Display Heat temp — Display current heating temperature (it is set temperature when enclosed by line)

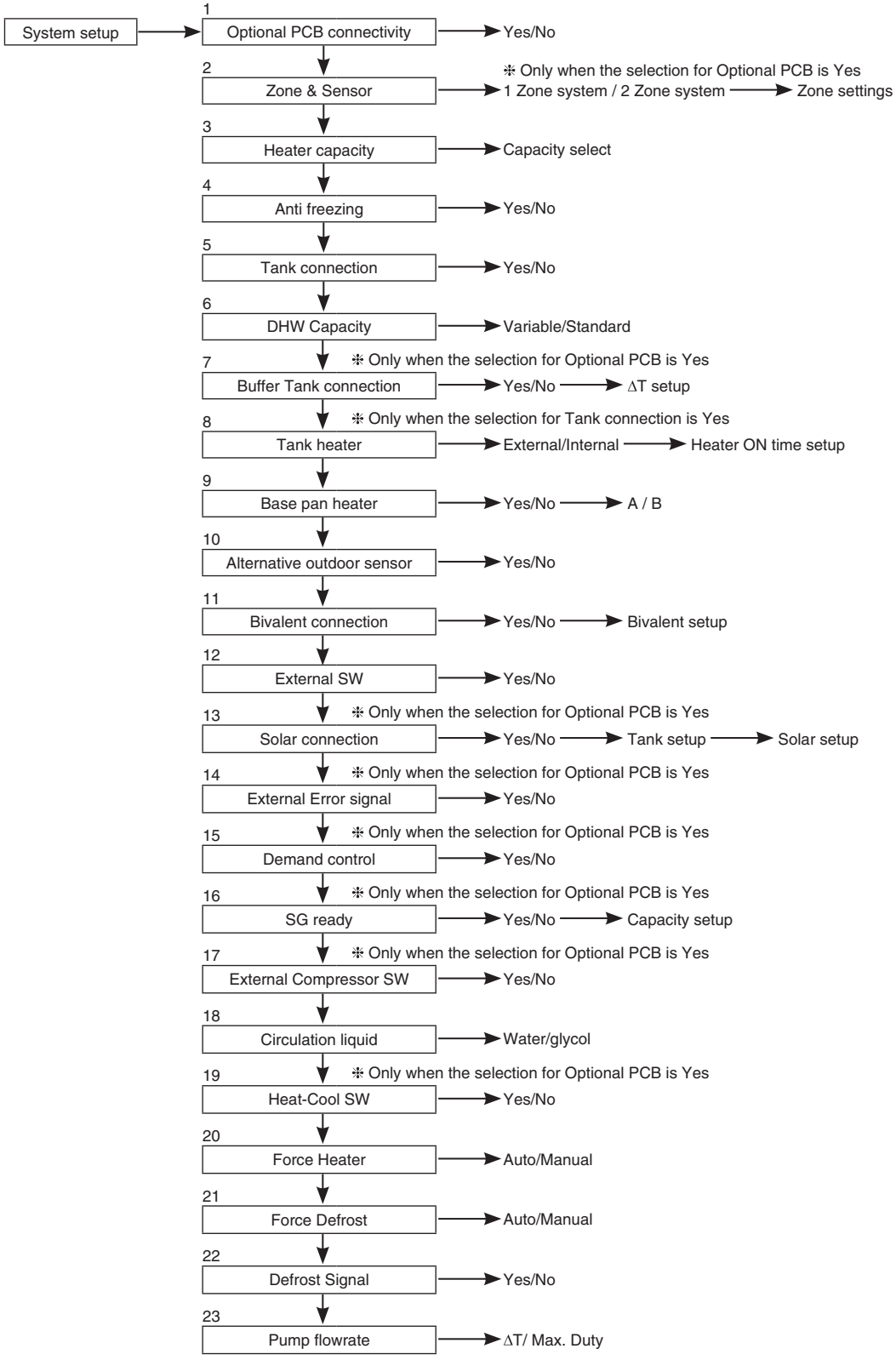
5: Display tank temp — Display current tank temperature (it is set temperature when enclosed by line)

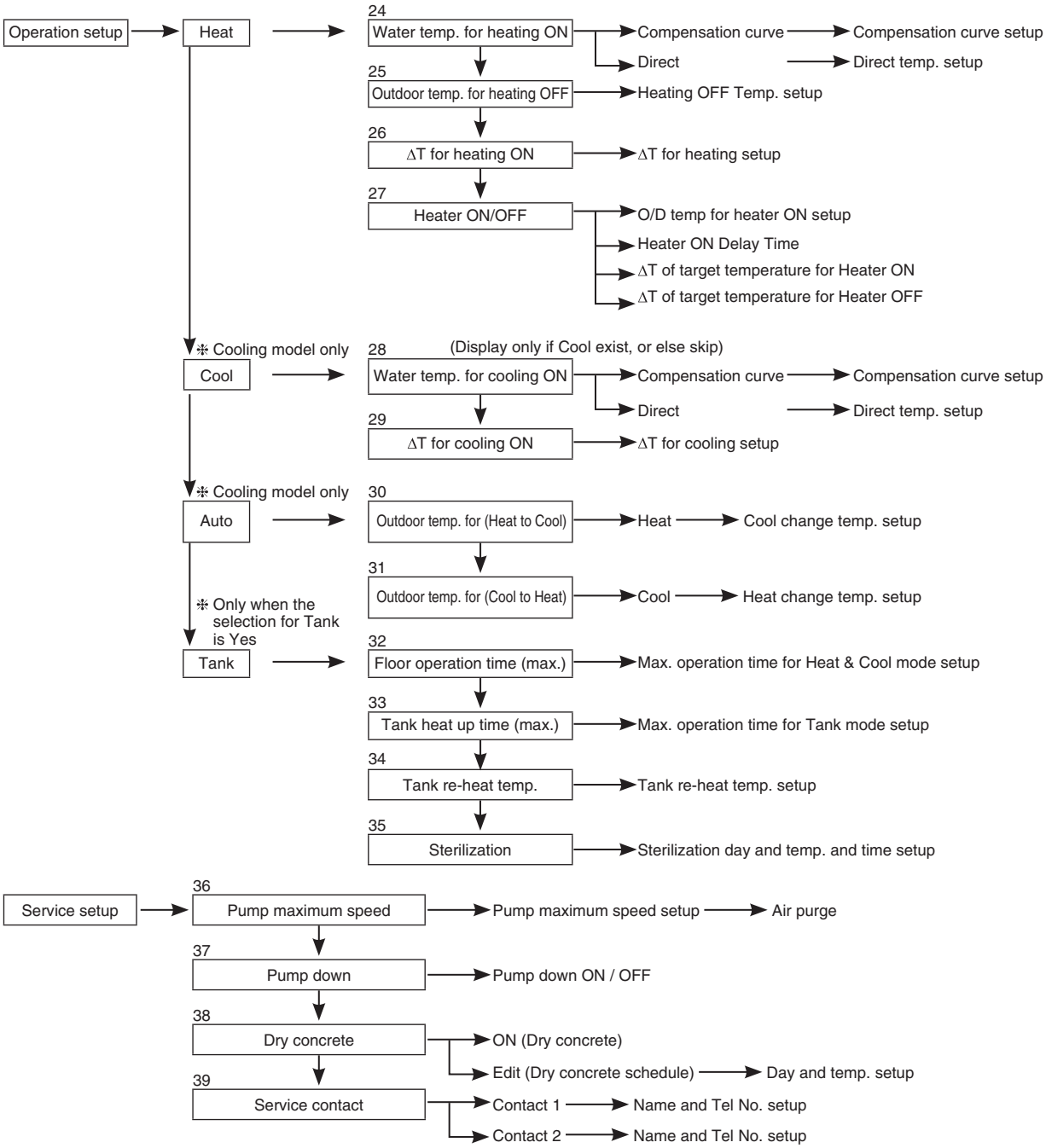
6: Outdoor temp — Display outdoor temp

First time of power ON (Start of installation)



12.4.2 Installer Setup



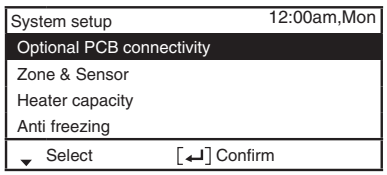

Operation setup
Heat
24
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
25
Outdoor temp. for heating OFF
Heating OFF Temp. setup
26
ΔT for heating ON
ΔT for heating setup
27
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❊ Cooling model only
Cool
28
(Display only if Cool exist, or else skip)
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
29
ΔT for cooling ON
ΔT for cooling setup
❊ Cooling model only
Auto
30
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
31
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❊ Only when the selection for Tank is Yes
Tank
32
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
33
Tank heat up time (max.)
Max. operation time for Tank mode setup
34
Tank re-heat temp.
Tank re-heat temp. setup
35
Sterilization
Sterilization day and temp. and time setup
Service setup
36
Pump maximum speed
Pump maximum speed setup
Air purge
37
Pump down
Pump down ON / OFF
38
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
39
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup

12.4.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install optional PCB.
Please select Yes after installing optional PCB.

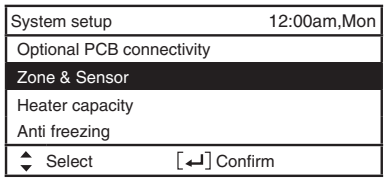
- 2-zone control
- Pool
- Buffer tank
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (Circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

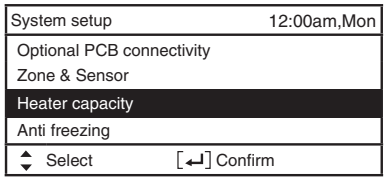
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

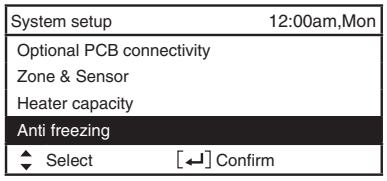
(NOTE) There are models which cannot select heater.



4. Anti freezing | Initial setting: Yes

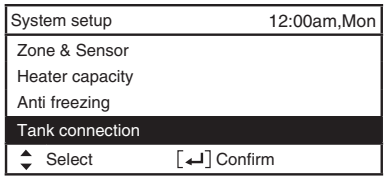
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



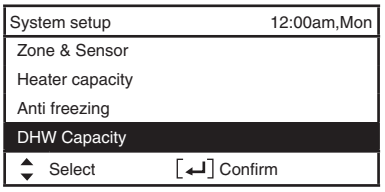
5. Tank connection | Initial setting: No

Select whether it is connected to hot water tank or not.
If set Yes, it becomes setting that uses hot water function.
Hot water temperature of tank can be set from main screen.



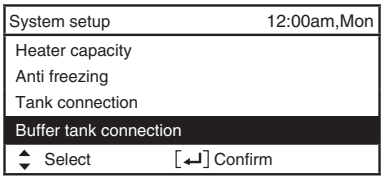
6. DHW Capacity | Initial setting: Variable

Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.



7. Buffer Tank connection | Initial setting: No

Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
(NOTE) Does not display if there is no Optional PCB.
If the buffer tank capacity is not so large, please set larger value for ΔT.

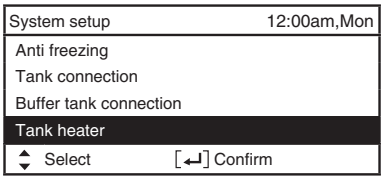


8. Tank heater | Initial setting: Internal

Select to use either built-in heater or external heater as heater for hot water tank.
If heater is installed on tank, please select External.

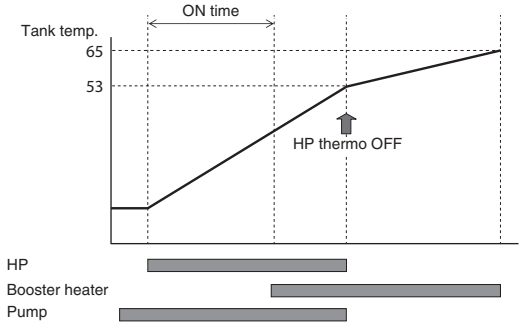
(NOTE) Does not display if there is no tank for hot water supply.

Please set “Tank heater” to “ON” in the “Function setup” from remote controller when using heater to boil the tank.



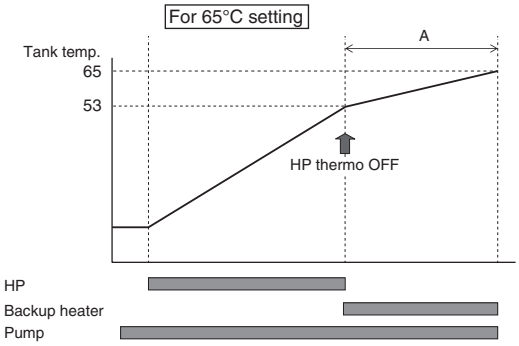
External A setting which is using booster heater installed on DHW tank to boil the tank.
The permissible heater capacity is 3kW and below.
The operation to boil the tank with heater is as below.
In addition, be sure to set suitable “Tank heater: ON time”

For 65°C setting



Internal A setting which is using backup heater of Mono bloc to boil the tank.
The operation to boil the tank with heater is as below.

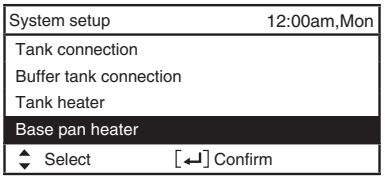
For 65°C setting



9. Base pan heater | Initial setting: No

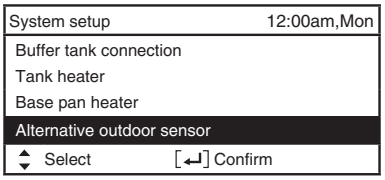
Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating



10. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.



11. Bivalent connection | Initial setting: No

System setup	12:00am,Mon
Tank heater	
Base pan heater	
Alternative outdoor sensor	
Bivalent connection	
Select	[↵] Confirm

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.

After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)

1) SG ready (Only available to set when Optional PCB set to YES)
   - SG Ready input from Optional PCB terminal control ON/OFF of boiler and heat pump as below condition

SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

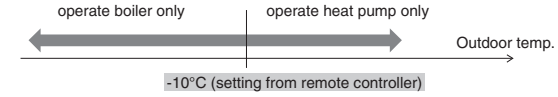
* This bivalent SG ready input is sharing same terminal as [16. SG ready] connection. Only one of these two setting can be set at the same time. When one is set, another setting will reset to not set.

2) Auto (If Optional PCB no Set, bivalent control pattern will set to this auto as default value)

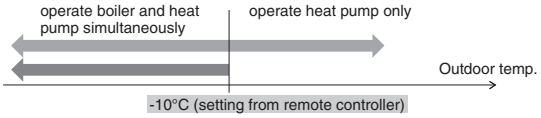
There are 3 different modes in the boiler operation. Movement of each modes are shown below.
① Alternative (switch to boiler operation when drops below setting temperature)
② Parallel (allow boiler operation when drops below setting temperature)
③ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is "ON", "boiler contact" is "ON", "_"(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

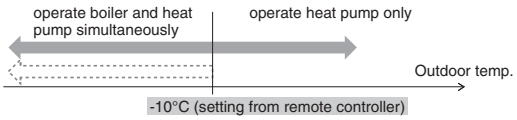


Parallel mode



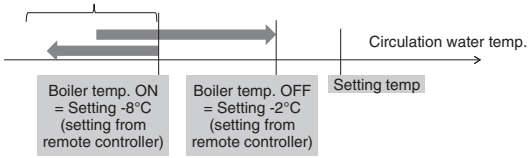
Advanced Parallel mode

For heating

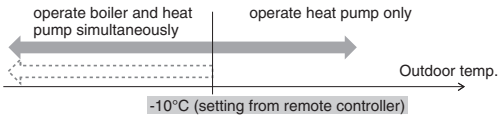


Although heat pump operates but water temperature does not reach this temperature for more than 30 mins (setting from remote controller)

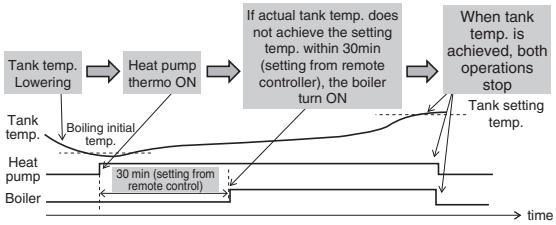
AND



For DHW tank



AND



In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of "Heating/Tank" mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

12. External SW | Initial setting: No

System setup	12:00am,Mon
Base pan heater	
Alternative outdoor sensor	
Bivalent connection	
External SW	
Select	[↵] Confirm

Able to turn ON/OFF the operation by external switch.